Evaluation of the 2023 Unemployment Insurance Reform in France

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April 9, 2026

Abstract

This paper evaluates the 2023 unemployment insurance reform in France, which introduced a state-dependent mechanism linking the duration of benefits to labor market conditions. During periods of favorable economic conditions, potential benefit duration is reduced, while full duration is restored when labor market indicators deteriorate. The reform was presented by the government as part of a broader strategy to promote full employment, strengthen work incentives, and reduce recruitment difficulties. The paper examines whether the institutional design of the reform is consistent with these objectives.

The analysis first situates the reform in the context of recent developments in the French labor market, notably the outward shift of the Beveridge curve and the coexistence of persistent unemployment with historically high vacancy rates. It then mobilizes search-and-matching theory to study the mechanisms through which changes in unemployment insurance duration may affect labor market outcomes. Particular attention is given to the role of labor market tightness, worker search behavior, wage determination, vacancy creation, and matching efficiency.

To analyze these mechanisms, the paper develops a steady-state search-and-matching model with time-limited unemployment benefits and endogenous search effort. The model identifies two central transmission channels of the reform: an incentive channel, through which shorter expected benefit duration increases insured workers' search effort before benefit exhaustion, and a bargaining channel, through which weaker outside options moderate wages and affect firms' vacancy-posting decisions. The framework is then calibrated to French labor market conditions in the second half of 2022 in order to simulate the effects of a 25% reduction in expected benefit duration. The analysis is further extended in two directions: first, by introducing wage rigidities through a binding minimum wage; second, by incorporating endogenous job destruction and business-cycle fluctuations in a multi-state macroeconomic environment.

Keywords: unemployment insurance, France, labor market tightness, Beveridge curve, search-and-matching models, benefit duration, recruitment difficulties, full employment

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1 Introduction

In 2023, France introduced a major reform of its unemployment insurance system by making the duration of benefits conditional on the state of the labor market. Under the new rules, potential benefit duration is reduced during periods of favorable economic conditions and restored when labor market conditions deteriorate. Presented by the government as part of a broader strategy to reach *full employment*, the reform was justified by a diagnosis that had become central to French labor market debates: the simultaneous coexistence of persistent unemployment and widespread recruitment difficulties. In this context, the reform was not framed merely as a budgetary adjustment, but as a structural instrument intended to improve matching between workers and firms by strengthening incentives to return to work.

This institutional change raises a broader economic question. Can a reduction in unemployment insurance generosity, and more specifically in benefit duration, effectively improve labor market performance when the economy displays both substantial unemployment and high vacancy rates? Standard job-search theory suggests that unemployment benefits affect search intensity, job acceptance behavior, and unemployment duration by modifying workers' outside options. Yet the aggregate consequences of these behavioral effects depend critically on the macroeconomic environment. In a slack labor market, stronger search incentives may simply redistribute a fixed number of jobs across job seekers, generating little effect on total employment. In a tight labor market, by contrast, where vacancies are abundant and firms face recruitment constraints, the same behavioral response may translate into genuine employment gains. The evaluation of the 2023 French reform therefore requires a framework capable of linking individual incentives to aggregate equilibrium outcomes.

This paper studies whether the design of the 2023 French unemployment insurance reform is consistent with the policy objectives assigned to it by the government. More precisely, it asks whether the reform can plausibly contribute to three stated goals: increasing employment, reinforcing work incentives, and reducing recruitment tensions. Answering this question requires going beyond the political rhetoric surrounding the reform and examining the mechanisms through which benefit duration can affect labor market equilibrium. The issue is not simply whether these objectives are desirable, but whether the specific institutional lever chosen by policymakers is well suited to achieve them in the French context.

The paper makes two contributions. First, it provides a theoretical and institutional interpretation of the reform by situating it within the recent transformation of the French labor market. We argue that the reform is best understood against the backdrop of a marked outward shift of the Beveridge curve in France, reflecting a deterioration in matching efficiency and a growing concern with labor shortages despite unemployment remaining above the levels observed in other advanced economies. This context

helps explain why the legislator adopted a state-dependent logic inspired by recent developments in the theory of optimal unemployment insurance, especially the idea that the optimal generosity of the system should vary with labor market tightness.

Second, the paper develops a sequence of search-and-matching models to identify and quantify the reform's transmission channels. We first build a steady-state model with time-limited unemployment benefits and endogenous search effort. This framework isolates two key mechanisms: an *incentive channel*, through which shorter expected benefit duration induces insured workers to intensify search before exhaustion, and a *bargaining channel*, through which weaker outside options moderate wages and encourage vacancy creation. We then calibrate the model to French labor market conditions in the second half of 2022 and simulate the reform as a 25% reduction in expected benefit duration. To assess the robustness of the results, we extend the baseline model in two directions. The first introduces wage rigidities through a binding minimum wage, allowing us to examine how the French institutional wage-setting environment may attenuate the reform's effects. The second introduces endogenous job destruction and aggregate productivity shocks in a multi-state business-cycle framework, allowing the reform to be evaluated across different macroeconomic regimes rather than under a single fixed steady state.

The results show that the reform is theoretically coherent with the objective of increasing employment in a relatively tight labor market, but that its effects are more limited and more ambiguous than the policy discourse suggests. In the baseline calibration, a reduction in benefit duration lowers steady-state unemployment through stronger search incentives and modest wage moderation. However, these aggregate gains remain quantitatively small relative to the broader objective of full employment. Moreover, the reform produces a compositional shift from insured to uninsured unemployment, revealing a trade-off between employment gains and income protection. The model also casts doubt on the claim that the reform substantially alleviates recruitment difficulties: vacancy creation reacts only weakly, and firms may even face slightly worse vacancy-filling conditions as the pool of effective job seekers contracts. When wage rigidity is introduced, the bargaining channel is largely shut down, and the reform operates mainly through worker-side incentives, further limiting its effect.

Taken together, these findings suggest that the reform's logic is not without foundation, but that its capacity to meet all three stated objectives simultaneously is partial and conditional. The reform appears better suited to strengthening search incentives than to resolving the structural matching problems underlying recruitment tensions. More broadly, the analysis highlights the importance of evaluating unemployment insurance reforms in general equilibrium and in relation to the business cycle, rather than inferring their effects from micro-level behavioral responses alone.

The remainder of the paper is organized as follows. Section 2 contextualizes the reform within recent

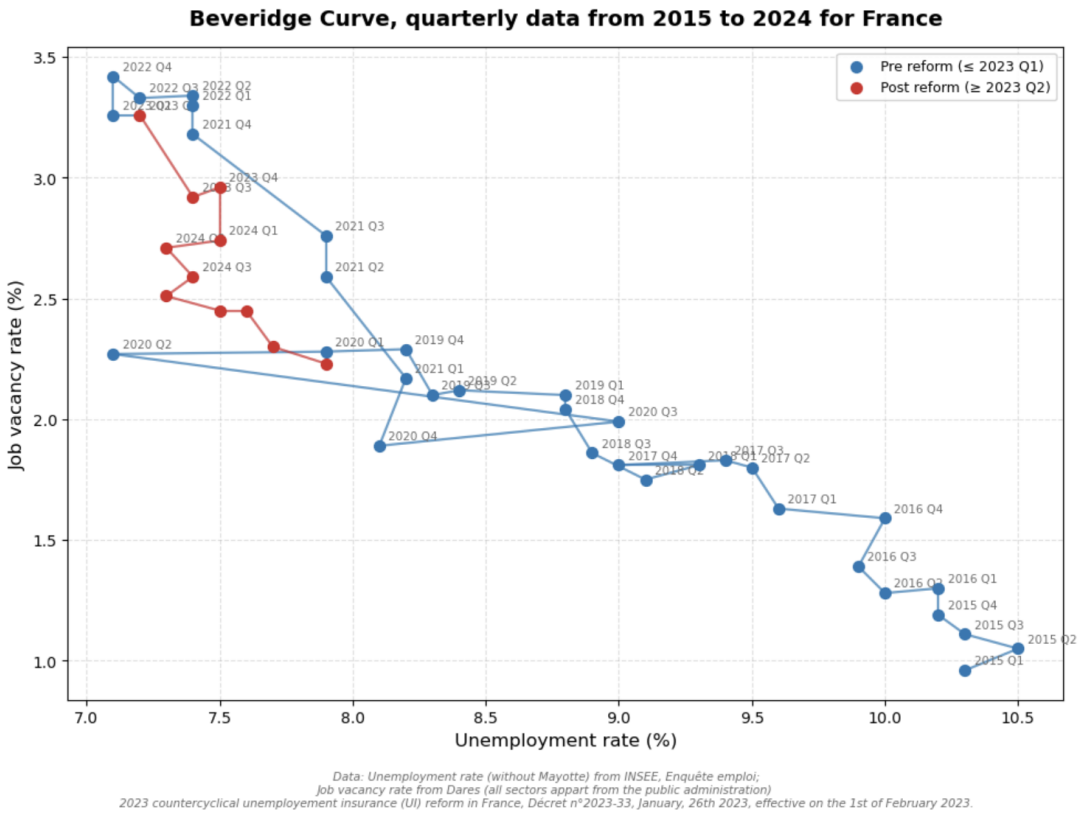
changes in the French labor market and the policy debate surrounding full employment, incentives, and recruitment difficulties. Section 3 explains why search-and-matching models provide an appropriate analytical framework for evaluating the reform. Section 4 reviews the relevant theoretical and empirical literature. Section 5 develops the baseline steady-state model and identifies the main transmission channels of the reform. Section 6 presents the calibration and reform simulation. Section 7 introduces a minimum-wage extension to analyze the role of wage rigidities. Section 8 extends the framework to endogenous job destruction and cyclical fluctuations. The final section concludes.

2 Labor Market Frictions and Institutional Adaptation: A Contextualization of the 2023 French Unemployment Insurance Reform

The 2023 Unemployment Insurance (UI) Reform in France established by Law n° 2022-1598 (21 December 2022) and Decree n° 2023-33 of (26 January 2023), introduced the principle of contracyclicity into the Labour Code. This reform mandates that the duration of benefits fluctuates according to economic indicators, tightening during periods of expansion, and expanding during downturns. This section aims to contextualize the reform by examining the "spirit of the law" - specifically why this change was enacted in 2023- and analyses the structural labor market tensions that lead to this shift.

2.1 The Beveridge Curve Shift in France: Labor Market Mismatches as a Catalyst for Reform

The 2023 UI reform in France was enacted in the context of a significant deformation of the Beveridge curve – the graphical representation of the inverse relationship between unemployment and job vacancies. Under standard theory, this relationship is negative: as unemployment falls, the pool of available workers shrinks, and vacancies naturally rise. However, recent evidence (2015–2022) for France suggests that the Beveridge curve became steeper or shifted outwards. This means that, for a given unemployment rate, the vacancy rate was higher than in the past. In other words, the observed decline in unemployment over recent years was accompanied by persistently high vacancy rates rather than a proportional reduction in job openings.



What makes the French case particularly striking is the threshold at which these tensions emerge. In economies like Germany, the United Kingdom, or the United States, steepened curves are observed when the labor market is at capacity, with unemployment rates fluctuating between 3% to 4%. In contrast, France began experiencing these high vacancy rates while unemployment remained comparatively high (around 7%). Such a configuration is generally interpreted as a decline in matching efficiency: the labor market becomes less effective at pairing job seekers with available positions. This macroeconomic diagnosis was corroborated by institutional data. By the final quarter of 2022, INSEE's quarterly business surveys revealed that recruitment tensions had reached unprecedented levels across all major sectors. Specifically, hiring difficulties were reported by 62% of firms in the services sector (excluding transport), 65% in manufacturing, and 81% in construction.

These figures represented historical peaks, signaling that the labor market had entered a zone of extreme tightness despite an unemployment rate that remained significantly above that of France's neighbors. While short-term labor supply and demand mismatches are often attributed to skill shortages, sectoral reallocation, or geographic frictions, the scale and synchronicity of these difficulties across diverse industries shifted the policy focus.

As recruitment tensions reached a breaking point, the political and academic debate increasingly turned toward the role of labor market institutions. Central to this discussion was the role of UI in

shaping individual behavior. From this perspective, the outward shift of the Beveridge curve might reflect how UI parameters influence the reservation wage and search intensity of the unemployed.

2.2 Unemployment Insurance and Matching Efficiency: A Behavioral Interpretation of the Beveridge Curve

One way to connect the coexistence of high vacancy rates and persistent unemployment is through the concept of moral hazard. In any insurance scheme, coverage may alter behavior: once individuals are insured against a risk, they may adjust their effort in ways that increase the likelihood or duration of the insured event. In the context of unemployment insurance, this concerns search effort and job acceptance decisions. This manifests in two primary ways:

- Reduced Search Intensity: benefits might allow individuals to search less intensively for new roles
- Increased Reservation Wages: job seekers might be more selective, rejecting offers that do not meet a certain wage threshold because the cost of remaining unemployed is subsidized.

Baily (1978) and Chetty (2006) provide a theoretical framework to understand moral hazard in the context of unemployment insurance. In its sufficient-statistics formulation, the optimal generosity of unemployment benefits balances two opposing forces: the welfare gain from consumption smoothing and the fiscal cost generated by behavioral responses. The key condition can be summarized as follows: the optimal level of benefits equates the marginal utility gain from raising consumption in unemployment with the marginal cost induced by the increase in unemployment duration. In practice, this trade-off depends on two sufficient statistics: (i) the drop in consumption upon unemployment, which measures the insurance value of benefits, and (ii) the elasticity of unemployment duration with respect to benefits, which captures the magnitude of moral hazard.

$$\frac{U'(c_u) - U'(c_e)}{U'(c_e)} = \frac{\varepsilon}{1-p}$$

This first-order condition characterizes the optimal generosity of unemployment benefits. The left-hand side captures the marginal insurance value of benefits, measured by the proportional gap in marginal utility between unemployment consumption c_u and employment consumption c_e. The right-hand side captures the marginal efficiency cost, where ε denotes the elasticity of unemployment duration with respect to benefits and p is the job-finding probability. At the optimum, the marginal insurance gain equals the marginal behavioral distortion.

Empirically, in the case of France, this behavioral response has been well documented. For example,

Iona Marinescu and Daphné Skandalis (2021) show that job search effort — measured by the number of applications sent — increases sharply as benefit exhaustion approaches, rising by about 50% shortly before benefits expire. Similarly, Le Barbanchon, Rathelot, and Roulet (2019) document a spike in exit rates from unemployment around benefit exhaustion. Specifically, the authors find that a 10% decrease in potential benefit duration leads to an average 3% decrease in unemployment duration.

2.3 When Does Moral Hazard Shift the Beveridge Curve?

In a slack labor market, defined by high unemployment and a scarcity of vacancies, the availability of jobs is "rationed." Under these conditions, the aggregate volume of hiring is primarily determined by labor demand rather than the intensity of job search. Consequently, increasing search incentives through benefit reductions is expected to trigger a displacement effect: while an individual may exit unemployment faster by searching more intensely, taking a position that would have otherwise been filled by another candidate. In this setting, moral hazard does not significantly alter the aggregate level of unemployment; it instead reshuffles the composition of the unemployment pool, redistributing a fixed number of job openings among a larger group of seekers.

This dynamic is inverted in a tight labor market. When vacancies are abundant and firms report systemic recruitment difficulties, the binding constraint on the economy shifts from a lack of demand to a lack of matching efficiency. In this state, the behavioral elasticities observed at the micro level—specifically how quickly a worker accepts a job or how intensely they search—have direct macroeconomic consequences. Because there is a surplus of available positions, an individual's successful search does not "displace" another worker; instead, it directly translates into aggregate hiring.

These distinctions between slack and tight labour markets and aggregate and individual behaviors are central to the framework developed by Landais, Michaillat, and Saez (2018). The core of this framework is an augmentation of the Baily-Chetty formula. While the standard model relies on micro-elasticity – measuring how an individual's search intensity responds to benefits – the LMS introduces macro-elasticity. In their work, the optimal UI (they express it as the replacement rate) is expressed as the sum of the Baily-Chetty term and a macroeconomic term.

$$R = \underbrace{\frac{u'(c_u) - u'(c_e)}{u'(c_e)} \cdot \frac{1-p}{\varepsilon}}_{\text{Baily–Chetty Term}}$$

This expression corresponds to the standard Baily–Chetty formula for optimal unemployment insurance. The proportional gap in marginal utility captures the marginal insurance value of benefits, while $\frac{1-p}{\varepsilon}$ scales this gain by the elasticity of unemployment duration with respect to benefit generosity. At the optimum, the marginal insurance benefit equals the marginal efficiency cost arising from search distortions.

This expression decomposes the optimal unemployment insurance formula into two components. The first term (Baily–Chetty term) captures the standard insurance–efficiency trade-off: the marginal insurance value of benefits, scaled by the behavioral elasticity of unemployment duration. The second term (elasticity wedge) reflects the gap between macro and micro elasticities and how labor market tightness $\Phi(\theta)$ amplifies or attenuates this distortion. When macro and micro elasticities coincide, the wedge disappears.

Building on this, the 2023 UI reform can be understood as a direct institutional application of the LMS framework. Within their model, the optimal generosity of the system is determined by the Elasticity Wedge (the ratio between macro-elasticity and micro-elasticity).

In the context of the post-2015 Beveridge curve shift, this theoretical framework provides the rationale for the reform's timing. When the labor market was slack, the macro-elasticity was significantly lower than the micro-elasticity, meaning search incentives were largely nullified by displacement effects. However, the steepening of the Beveridge curve and the spike in recruitment tensions signaled that the macro-elasticity had converged toward the micro-elasticity. In this state, the wedge closes: search externalities vanish, and the social cost of moral hazard rises because individual effort now translates directly into aggregate hiring.

While the LMS framework formally identifies the replacement rate as the optimal variable for adjustment, the 2023 reform utilizes their state-dependent logic to modulate another UI parameter: benefit duration. Although the specific parameter differs, the underlying economic principle remains identical: the generosity of the system must be indexed to the prevailing macroeconomic regime to balance insurance with matching efficiency. What follows is a discussion on why duration was selected as the primary lever for this institutional adaptation in the French context.

2.4 Dealing with Moral Hazard in Unemployment Insurance: Level vs. Duration

Having established that moral hazard significantly impacts aggregate employment in a tight labor market, a critical policy question remains. If the macroeconomic context justifies a reduction in UI generosity to improve matching efficiency, which specific parameter of the system should be adjusted? While LMS (2018) framework identifies the replacement rate (the benefit level) as the primary variable for adjustment, policymakers in France can regulate:

- Replacement rate (benefit level relative to past wages)
- Potential duration of benefits
- Eligibility conditions

Historically, France's primary lever for modulating UI system has been eligibility conditions rather than benefit levels. To manage the system's solvency, successive governments typically adjusted the affiliation threshold—the minimum period of employment required to qualify for benefits. For instance, the 2019 reform increased this requirement from 4 months of work over the previous 28 months to 6 months within a 24-month window.

Furthermore, the French system has historically resisted changes to the replacement rate due to its "contributory" nature and significant social opposition. The most notable attempt to modulate benefit levels occurred in 1992 with the introduction of degressivity (a sliding scale where benefits decreased every six months). This measure was widely condemned as socially punitive and was abolished in 2001.

The 2023 French reform primarily targeted potential benefit duration. However, alternative instruments could also have been used, and each carries distinct implications for redistribution and work incentives.

The theoretical literature emphasizes that the specific design of UI matters as much as its generosity. Reducing benefit levels might significantly harm liquidity-constrained individuals, particularly low-income workers who rely on benefits to smooth consumption. Building on the Baily-Chetty sufficient-statistics framework, Bonnet et al. (2024) estimated the consumption drop upon job loss using high-frequency French bank account data. They found that 36% of income loss is immediately offset by spending cuts (an MPC of 0.35), a figure that rises to 46% after six months as liquid savings are depleted. Because this marginal propensity to consume is significantly higher for low-income households, cutting benefit generosity would be at the origins of a substantial welfare loss.

By contrast, adjusting potential duration rather than the replacement rate may preserve income

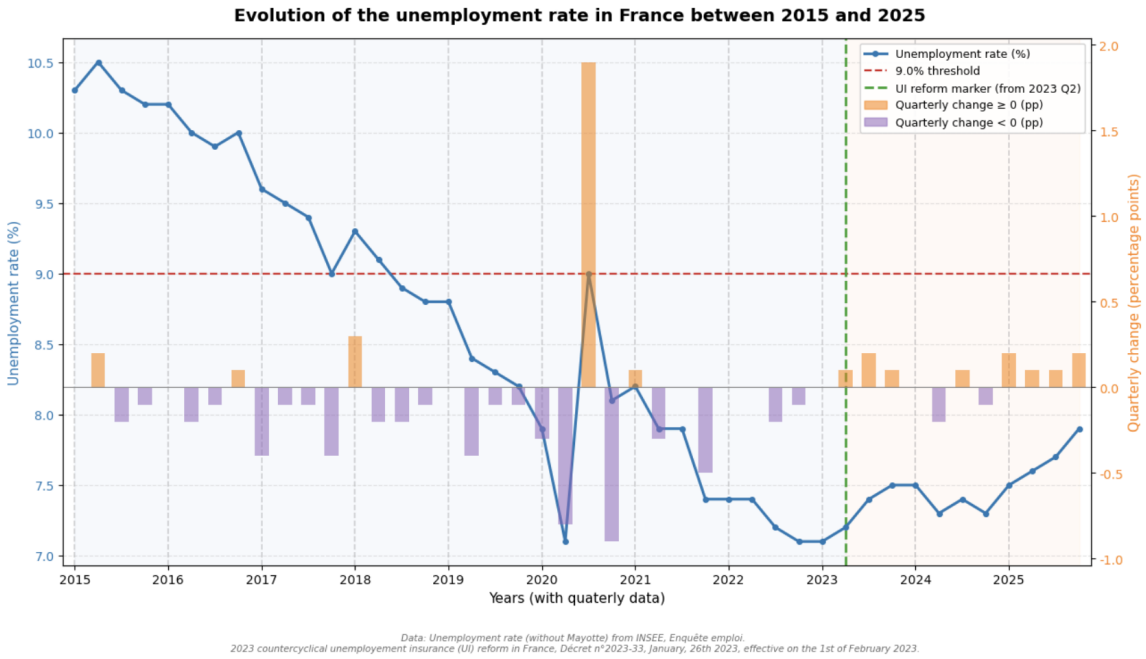
support in the early phase of unemployment — when liquidity needs are strongest — while increasing incentives as the spell lengthens. In that sense, the 2023 reform can be interpreted as targeting the timing of incentives rather than the generosity of short-term insurance. Moreover, because the reform applies uniformly across workers, it avoids explicitly differentiating across income groups. From a salience perspective, adjusting potential benefit duration is particularly effective because the “end-of-benefit" data acts as a powerful psychological anchor.

2.5 Measuring Tightness: The Selection of Macroeconomic Thresholds for Benefit Modulation

The legislator selected two specific macroeconomic indicators to serve as objective proxies for labor market tightness.

- The Structural Threshold: The 9% Unemployment Rule The 25% reduction in benefit duration is conditioned on a national unemployment rate remaining below 9%. Above this level, the market is assumed to be in a state of job rationing.
- The Momentum Trigger: The 0.8 Percentage Point Rule This criterion is intended to detect sudden deterioration within a single quarter. Even if the overall unemployment rate remains relatively low, an increase of 0.8 percentage points or more over a quarter may signal a rapid shift toward a more constrained, rationed labor market environment.

The robustness of these thresholds is evidenced by a retrospective simulation of the French labor market over the last two decades.



Had these state-dependent rules been implemented in 2003, the interaction of the structural and momentum triggers would have accurately identified the two most significant macroeconomic shocks of the era:

- The 2009 Global Financial Crisis: The 0.8pt trigger would have activated full duration protections at the onset of the crisis, which would then have been maintained by the structural 9% rule until 2018.
- The 2020 COVID-19 Shock: The sudden spike in unemployment would have triggered an immediate restoration of full benefits.

3 Policy Objectives and Research Question

Evaluating a public policy reform first requires clarifying the objectives it was intended to achieve and the mechanisms through which it is expected to attain them. In the case of the 2023 French unemployment insurance reform, the government presented the reform as part of a broader strategy aimed at achieving **full employment** (*le plein emploi*) in a labour market that policymakers themselves described as "paradoxical," combining persistent unemployment with widespread recruitment difficulties.

In his address to the National Assembly on September 13, 2022, the Minister of Labour, Olivier Dussopt (2), presented the reform as a policy instrument intended to serve three main objectives: contributing to full employment, strengthening work incentives, and reducing recruitment difficulties. Taken together, these objectives imply a common causal claim: that changing unemployment insurance rules can improve labour market performance by accelerating transitions into employment and easing firms' hiring constraints.

3.1 Contributing to the Objective of Full Employment

First, the reform was framed as part of a broader governmental strategy aimed at reaching full employment. In the Minister's words:

> *"It [the roadmap] is dominated by one objective: full employment. Beyond a campaign commitment, it is the course that the Government wishes to give to all the policies it carries out... full employment is certainly an economic objective, but also a political and social one."*

Within this broader strategy, unemployment insurance reform was presented as one policy lever among others designed to improve labour market functioning and accelerate returns to employment.

What Does "Full Employment" Mean in the French Context?

In the French policy debate, *full employment* does not mean the complete disappearance of unemployment. Rather, it refers to a situation in which unemployment is reduced to a residual level corresponding mainly to transitional or frictional unemployment, that is, the time required for workers to move between jobs or enter the labour market. French public sources explicitly note that full employment is therefore compatible with a non-zero unemployment rate and does not imply the absence of labour market adjustment frictions.

In recent French policy discourse, this notion has generally been operationalized as a unemployment rate below approximately 5% in the sense of the International Labour Organization (ILO/BIT). This threshold has become the benchmark commonly used by policymakers to define the objective of *plein emploi*, including in the debate surrounding the 2023 unemployment insurance reform and the 2023 law for full employment.

However, this benchmark should be interpreted with caution. First, full employment remains a convention rather than a mechanical threshold. Second, it does not capture underemployment, job quality, or the extent of mismatch between workers and vacancies. Reaching a 5% unemployment rate would therefore not necessarily imply that all labour market frictions have disappeared, but rather that unemployment has been reduced to a level considered close to its irreducible minimum in the French context.

3.2 Strengthening Work Incentives

Second, the reform was intended to reinforce incentives to return to work during favourable economic conditions. The government argued that the previous system did not sufficiently adjust to the business cycle and could weaken job-search incentives when labour demand was strong.

> *"When the economic situation improves, [the current model] is too protective and it loses its incentive character (il fait perdre tout caractère incitatif)… when the economy is good, the rules must be more incentive-based."*

This argument rests on the idea that unemployment insurance generosity affects search behaviour and job acceptance decisions, and that these behavioural responses should be made more state-dependent.

3.3 Reducing Recruitment Difficulties

Third, the reform was also presented as a response to the coexistence of persistent unemployment and widespread recruitment difficulties reported by firms.

"During periods of job creation, the share of unemployed individuals eligible for unemployment benefits exceeds 60%, whereas it falls to around 45% during economic downturns. Our system is therefore paradoxical. At the same time, the number of vacant jobs increased by a factor of 2.5 between 2017 and 2022, while the number of unfilled job vacancies per 1,000 registered job seekers rose from 50 to 170. These figures confirm the recruitment difficulties reported by many business leaders: 60% of them report such difficulties, and 30% of industrial firms state that their production capacity is constrained as a result."

The reform was thus framed not only as an instrument for reducing unemployment, but also as a mechanism for improving the matching of job seekers to available vacancies. Implicit in this argument is the claim that tightening unemployment insurance rules can alleviate firms' hiring difficulties by accelerating labour reallocation.

3.4 A Parliamentary Counterpoint

These objectives, however, were contested during parliamentary debates. In particular, Deputy Olivier Serva (LIOT) challenged the underlying logic of the reform in a single intervention that questioned the empirical and causal foundations of all three objectives:

"This minimal text, which claims to address recruitment difficulties, in reality deals with unemployment insurance—as if the link between the two were obvious and automatic [...] recruitment difficulties are multifactorial [...] our benefit system already encourages a return to work. Why tighten it further? The last reform came into force barely a year ago. Do we really have enough hindsight?"

Serva's intervention is analytically important because it does not merely oppose the reform politically; it challenges the mechanisms through which the government expects the reform to operate. Three objections follow.

First, with respect to full employment, Serva questions whether sufficient time has elapsed since the previous reform to justify additional tightening, suggesting that the empirical basis for expecting further gains in employment remains uncertain.

Second, with respect to work incentives, he argues that the existing system already encourages return to employment, directly contesting the government's claim that current rules are insufficiently incentive-compatible.

Third, with respect to recruitment difficulties, he challenges the assumption that unemployment insurance is the appropriate policy lever. His argument that only 40% of registered job seekers receive

benefits and that recruitment tensions are "multifactorial" casts doubt on any simple or automatic link between tighter benefit rules and improved hiring outcomes.

This final objection is especially important for the analysis that follows. If recruitment difficulties reflect not only workers' incentives but also the characteristics of the jobs offered, the efficiency of the matching process, or broader frictions between vacancies and job seekers, then the effects of unemployment insurance reform cannot be inferred from intuition alone. They must be analysed within a framework capable of identifying the conditions under which changes in benefit duration affect search effort, job acceptance, vacancy creation, and ultimately hiring outcomes.

3.5 Research Question

The policy debate therefore reveals that the reform is not simply associated with a set of stated objectives; it also rests on a set of contested causal claims. The question is not only whether full employment, stronger incentives, and reduced recruitment difficulties are desirable goals, but whether the specific institutional design of the 2023 reform is capable of advancing them.

This leads to the central research question of the dissertation:

> *To what extent is the design of the 2023 French unemployment insurance reform consistent with the policy objectives articulated by the government?*

To answer this question, the analysis focuses on three specific dimensions:

1. **Full Employment:** To what extent can a reduction in benefit duration contribute to increasing employment?
2. **Incentives:** Is the contracyclical modulation of unemployment insurance sufficiently calibrated to strengthen job-search incentives during periods of economic expansion?
3. **Recruitment Tensions:** To what extent can the institutional changes introduced by the reform alleviate the recruitment difficulties reported by firms?

Addressing these questions requires a framework capable of identifying the mechanisms through which unemployment insurance rules influence labour market outcomes. The next section therefore introduces the analytical framework used in this study and explains why search-and-matching models are particularly well suited to analyse the transmission from unemployment insurance policies to job search behaviour, vacancy creation, and unemployment dynamics.

4 Why Search-and-Matching Models Are Well Suited to Analyze the 2023 Reform

In his Nobel Prize lecture, Pissarides (11) emphasizes that unemployment in modern economies arises not primarily from wage rigidity, but from search frictions that prevent instantaneous coordination between workers and firms. Search-and-matching models developed by Diamond (1), Mortensen (7), and later formalized in Mortensen and Pissarides (8), model labor markets as environments where matching between workers and firms is costly and time-consuming. Because matching requires time, information, and resources, unemployment and vacancies coexist in equilibrium. This characterization closely describes the French labor market prior to the reform, where, as detailed in the previous section, elevated recruitment difficulties persisted despite significant unemployment.

Limitations of the neo-classical Walrasian Framework for Evaluating the 2023 Reform

In a Walrasian labor market, employment is determined by the intersection of an upward-sloping labor supply curve and a downward-sloping labor demand curve, with wages adjusting to clear the market. In this framework, the labor market operates as a spot market in which workers and firms can instantaneously form employment relationships at the equilibrium wage. Unemployment, if present, is therefore voluntary or reflects wage rigidities or institutional constraints rather than coordination frictions between workers and firms (14). As a result, changes in unemployment insurance primarily affect labor supply decisions, with limited implications for firms' hiring behavior beyond standard demand responses to wage adjustments. This structure makes it difficult to account for the persistent coexistence of unemployment and recruitment difficulties observed in France prior to the 2023 reform.

A search-and-matching framework is therefore particularly well suited to analyze the reform for three fundamental reasons.

- First, it provides a coherent explanation for the coexistence of unemployment and vacancies observed in France, interpreting the Beveridge curve shift as an equilibrium outcome of changes in matching efficiency.
- Second, it integrates individual search incentives — such as reservation wages and job acceptance decisions — with aggregate employment outcomes, thereby allowing micro-level behavioral responses to influence macroeconomic equilibrium.
- Third, it permits policy effects to vary across macroeconomic regimes, a feature that is essential for assessing a reform explicitly designed to be state-dependent.

The following section outlines the core ingredients of the search-and-matching framework and explains why each of them is directly relevant for evaluating the 2023 reform.

4.1 Matching Function: Accounting for the Paradox of Concurrent Vacancies and Unemployment

At the core of the search-and-matching framework lies a matching function:

$$M(U, V),$$

where U denotes unemployment and V denotes vacancies. Hiring depends on both sides of the market and on the efficiency with which workers and firms meet.

Relevance for Evaluating the 2023 Reform

The matching function is a central ingredient of search-and-matching models because it captures a fundamental feature of modern labor markets: employment is not created automatically when unemployed workers and vacancies coexist. Instead, hiring results from a coordination process that depends simultaneously on the number of job seekers, the number of posted vacancies, and the efficiency with which the two sides meet. This representation is particularly relevant for analyzing the French context prior to the 2023 reform, where as explained above, significant unemployment coexisted with historically high recruitment difficulties.

Importantly, it also provides a framework to study how unemployment insurance policy parameters—such as benefit duration—can influence aggregate employment outcomes. By affecting search intensity or job acceptance behavior, unemployment insurance alters the rate at which matches are formed, thereby shaping both unemployment dynamics and firms' hiring conditions.

4.2 Labor Market Tightness : A Proxy to the Reform's State-Dependent Logic

A central state variable in the search-and-matching framework is labor market tightness:

$$\theta = \frac{V}{U}.$$

Relevance for Evaluating the 2023 Reform

Tightness captures the balance between vacancies and unemployment, and therefore the extent to which firms face hiring constraints. Since the 2023 reform explicitly conditions benefit duration on the unemployment rate—an institutional proxy for labor market tightness—its macroeconomic impact cannot be assessed independently of this state variable

4.3 Vacancy Creation Dynamics: Evaluating the Reform's Impact on Firm-Side Incentives

Another defining feature of the Mortensen–Pissarides model is endogenous vacancy creation. In these models, employment depends not only on how quickly workers accept jobs, but also on whether firms choose to create vacancies in the first place

Relevance for Evaluating the 2023 Reform

In search-and-matching frameworks, firms actively decide whether to create vacancies by comparing the expected profitability of hiring with the cost of posting a vacancy. Unlike in Walrasian models—where firms simply hire labor at the market-clearing wage and do not make explicit vacancy-posting decisions—firms in frictional labor markets respond strategically to labor market conditions and policies. This feature is particularly important for studying the 2023 unemployment insurance reform. By changing the duration of benefits, the reform can affect job search intensity and reservation wages, which in turn influence firms' vacancy-posting decisions. Incorporating endogenous job creation allows the reform to impact both sides of the labor market—job seekers and firms.

4.4 The Bargaining Channel: Quantifying UI-Induced Shifts in Worker Outside Options

In search-and-matching models, wages are not exogenously given nor do they simply adjust to clear the labor market as they do in Walrasian models. Instead, they are determined through a bargaining process between workers and firms, reflecting the surplus generated by a match and the outside options available to each party.

Relevance for Evaluating the 2023 Reform

Wage determination plays a central role in transmitting the effects of unemployment insurance to labor market outcomes. In frictional labor markets, unemployment benefits shape workers' outside options: more generous or longer-lasting benefits increase the value of remaining unemployed and strengthen workers' bargaining power in wage negotiations. Higher negotiated wages reduce the surplus from a match and may discourage vacancy creation, while shorter benefit duration can weaken outside options and increase firms' incentives to post vacancies. This channel is particularly relevant in the French context, where wage-setting institutions and relatively limited wage flexibility may amplify or attenuate the general equilibrium effects of unemployment insurance reforms.

4.5 Search Intensity: Modeling the Behavioral Transition to Accelerated Re-employment

In search-and-matching models with endogenous search effort, unemployed workers actively decide how intensively to search for jobs. Search effort influences the probability of receiving job offers and therefore plays a key role in determining unemployment duration and the speed at which vacancies are filled.

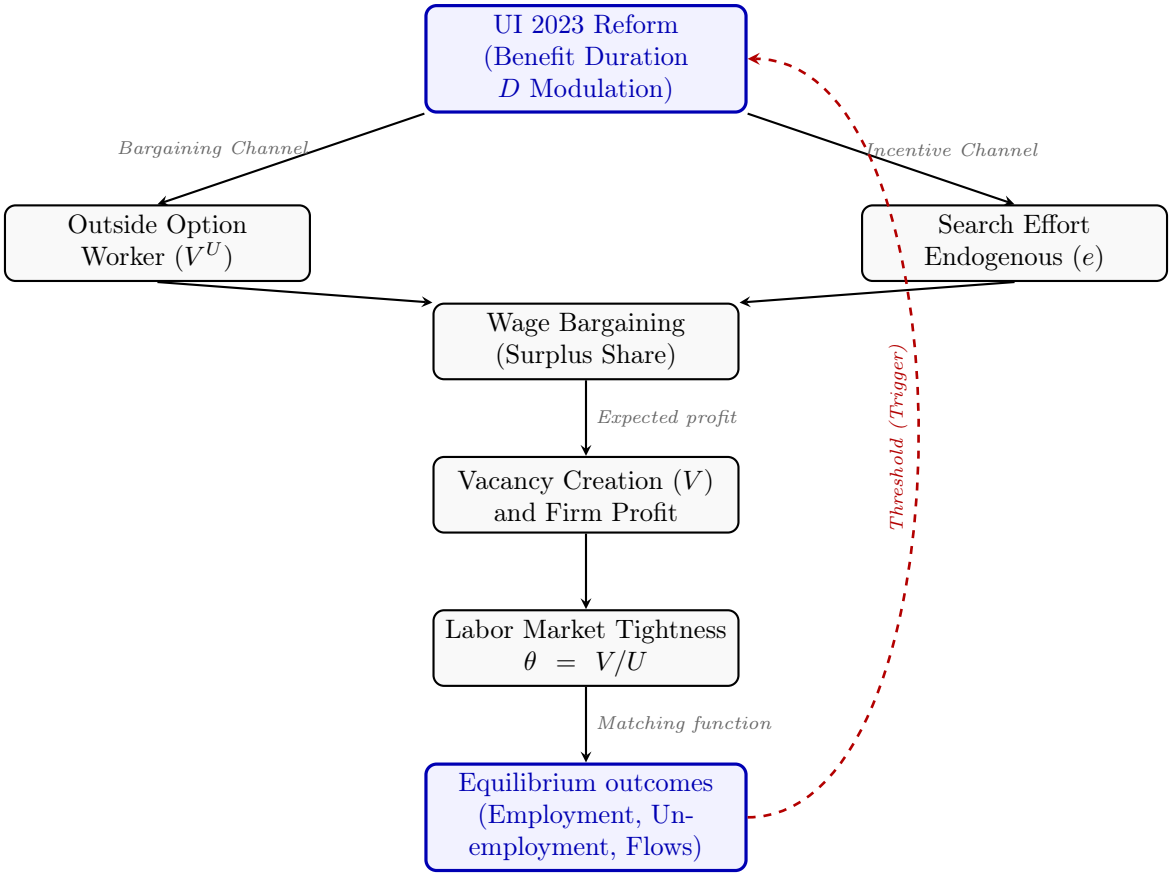


Figure 1: The General Equilibrium Transmission of the 2023 Reform. The diagram illustrates how benefit duration affects both the worker's bargaining position and search intensity, which jointly determine firm entry and market tightness, ultimately looping back to trigger the reform's modulation logic.

4.6 Evolution of search and matching publications, as references of unemployment insurance models

The relevance of search-and-matching models for analyzing unemployment insurance schemes is reflected in the positive evolution of the academic works using such frameworks. It is presently displayed in the frequency graph presented hereafter, Figure 2, depicting the annual count of academic publications using search-and-matching frameworks in unemployment insurance works. It has been

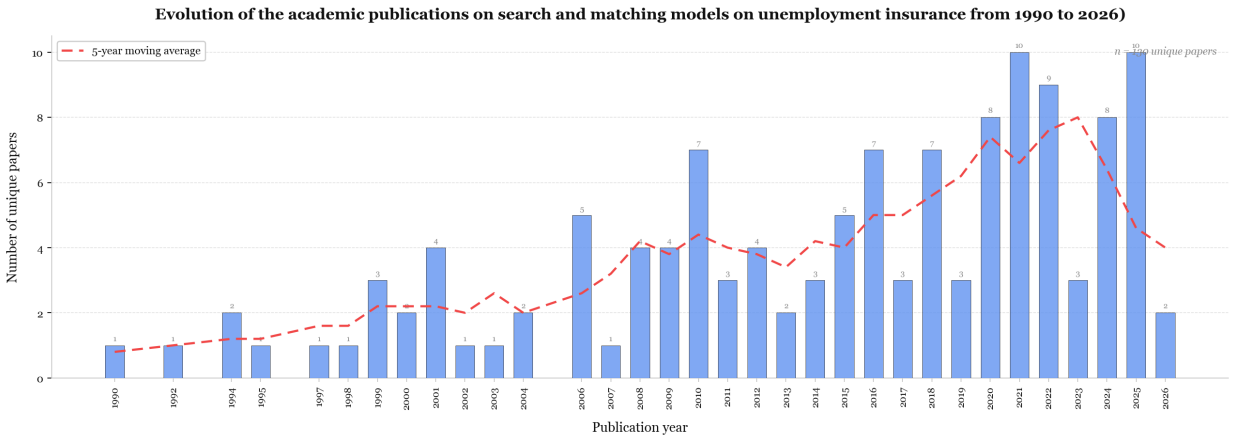


Figure 2: Annual count of unique academic publications of **scrapped** search-and-matching models from unemployment insurance academic publications from 1990 to 2026 Source: Google Scholar, as of March 2026 (though automated scraping).

constructed from the web scrapping of Google Scholar from 1990 to 2026.[1]

The publication distribution reveals three major phases. In a first instance, more or less until the mid 2000s, the literature was quite sparse. This period corresponds to the consolidation of the theoretical foundations of search-and-matching models: initially developed by Diamond (1) and progressively consolidated through papers such as Mortensen (7), that are seen as references, the framework is later formalized in Mortensen and Pissarides (8).

An acceleration emerges after that. It reflects both the democratization of the framework and the necessary refinement, as numerous extensions of the model are continoulsy proposed. This craze might has simultaneously been enhanced with a growing interest for the significant social implications of UI policies, and by their fundamental role in shaping labor market conditions.
A second and more sustained acceleration seems to emerge around the year 2020, with two publication peaks in 2021 and 2025. This recent surge likely reflects renewed attention for unemployment insurance designs worlwide following the Covid 19 crisis. During which unprecedented extensions of benefit schemes highlighted the crucial role of UI policy parameters and job-search behaviors.

Overall, the moving average line confirms a clear upward trend. This pattern illustrates both the maturation of the search-and-matching frameworks as an important and relevant tool for analyzing frictional labor markets.The present analysis hence builds on a growing and an 'up to date' model of the UI literature, while targeting the rigid French institutional setting: *l'assurance chômage*, that has received comparatively less attention than Anglo-Saxon unemployment insurance systems.

[1]Methodological note: papers were collected through automated queries scrapping on Google Scholar. Duplicates are removed whenever the title similarity exceeded 93%, and lnly papers whose title or abstract simultaneously contained explicitly "unemployment insurance" and a reference to search-and-matching terminology were retained.

5 Literature review

Foreword: The present evaluation project is deemed to highlight key macro mechanism to seize the implications of the 2023 reform in France.

It thus appears important to emphasize that, overall, the macroeconomic literature on unemployment insurance is much scarcer than the micro literature. The vast majority of existing studies focus on micro dynamics, partly due to the difficulty of analyzing long term effects with sufficient precision. In addition, external changes are likely to affect labor market outcomes such as legislative changes, human capital depreciation or changes in job compositions and quality. The following literature review aims at presenting a combination of relevant micro and macro contributions that have supported our research, the understanding and quantification of core mechanisms underlying the French reform.

The first strand presents how unemployment insurance affects search incentives and job finding behaviors, while the second strand focuses on how unemployment insurance affects wage setting dynamics through vacancy posting and workers outside options; affecting matching efficiency, labor-market tightness, and equilibrium unemployment.
The above mentioned dimensions have been described for the French environement, prior to the studied reform: Cahuc, Carcillo, and Landais (2021)(14) provide the rationale behind the institutional debate by exposing a French pre reform system as poorly adapted to cyclical labor market conditions and insufficiently coordinated. Their call for policy changes links unemployment insurance reform to both: work incentives and recruitment difficulties. The introduction of counter cyclical benefits emerges as a possible policy lever to ensure efficient responses from the national UI system to business cycles variations.

5.1 Search incentives and job finding behaviors, evidence from the French context

Incentive effects of unemployment insurance on individual search behaviors are historically studied through the standard benchmark Mortensen (1977)(21) search and matching framework. Unemployment insurance hereby improves individual insurance against income loss but also weakens incentives to search and accept jobs quickly. In this tradition, Baily (1978)(1) and, later, Chetty (2008)(6) formalize the basic trade-off between consumption smoothing and moral hazard in the literature. More generous

insurance schemes raise welfare by protecting consumption, but it can also increase unemployment duration if it reduces search effort or raises workers selectivity, as reservation wages mechanically increase for the insured.

For France, Marinescu and Skandalis (2019)(18), document changes in search timings in response to UI extension using a search and matching model based on Mortensen model (1997). To match national settings, it is enhanced with finite entitlements duration and selective eligibility criterion. Workers are found to significantly increase their search intensity shortly before benefits exhaustion (quantified as a spike of approximately 60 percents increase) while target wages simultaneously decrease as exhaustion approaches. This mechanic shows that potential benefit duration qualitatively affects re employment dynamics even if the depicted adjustment occurs late in the spell.
Standard search models are in this sense useful for interpreting the effect of unemployment insurance on job search behavior. It frames that more generous schemes mostly delay the pre exhaustion surge, hence depicting potential benefit duration as a key lever for policy makers. To build on this idea, Le Barbanchon, Rathelot, and Roulet (2017)(16) exploit French administrative data to estimate how potential benefit duration affects reservation wages and job selectivity. As they distinctly focus on selectivity, they quite surprisingly find that reservation wages weakly respond to longer benefit duration[2]. They nonetheless report that they cannot reject a zero elasticity of reservation wages with respect to potential benefit duration. This matters in the interpretation of moral hazard as the evidence suggests that the main adjustment passes through the timing and intensity of search rather than through large changes in reservation wages. French evidence points out to delayed search over an upward shift in wage expectations.

These micro evidences contradict the intuition that UI eligible workers demand substantially higher wages throughout unemployment while being covered the programs. It depicts a more nuanced mechanism: benefit duration affects search intensify and decreased selectivity close to benefits exhaustion. We find thus interpretation to be consistent with the counter cyclical modulation introduced in France through the 2023 reform.

For tight labor markets, those facing difficulties to fill-in vacancies, as nationally translated by the steepening of the French Beveridge curve; behavioral changes in search timing can be significant for aggregate matching outcomes. Landais, Michaillat, and Saez (2018)(14) nuance the aggregate consequences of search incentives by highlighting their strong correlation with labor market conditions.

[2]An additional month of benefits for a claimant entitled with 10 months of potential benefits duration increases the reservation wage by 0.06%

Micro responses might not always translate into aggregate employment. As explained in section 2.3, in slack labor markets, stronger search effort may mostly reallocate jobs. To go further in the analysis, they additionally incorporate equilibrium labor market tightness[3] in their welfare analysis. In their framework, unemployment insurance not only influences search incentives but also the equilibrium ratio between vacancies and job seekers in tight markets. They show that the optimal UI Baily–Chetty formula is to be complemented with a macro contra cyclical correction term. The idea behind this correction term is that when tightness is inefficiently low, notably during recessions and periods of low labor demand; more generous UI can improve welfare as it increases workers outside option as well as the vacancy equilibrium through firms posting decisions, and that despite moral hazard. Reversely, when tightness is already high, stronger search incentives may have larger employment effects. This insight implies that unemployment insurance policy should account for how benefit levels influence both worker incentives and the equilibrium conditions governing job creation and matching efficiency. Thereby making the optimal policy counter cyclical.

5.2 Wage setting, vacancy posting, outside options, and equilibrium unemployment

The second strand of the literature moves from search responses to labor market adjustments. The idea being that unemployment insurance not only affects workers behavior but also firms, thereby transcribing larger macro effects. In search and matching models with wage bargaining, UI raises the value of the outside options which translate into higher negotiated wages, lower expected profits from a match for the firm, thus, reduced vacancy creation. Fredriksson and Holmlund (1998)(9) provide an early foundational analyses of this mechanism. In their equilibrium search and matching model, unemployment insurance affects outcomes through both endogenous search effort and wage bargaining, under a free entry condition. The wage channel is found to be quantitatively important: more generous unemployment benefits strengthen workers bargaining positions, raises negotiated wages, discourages low value vacancy creation, and increases equilibrium unemployment. In addition, they depict the optimal insurance schedule with finite declining benefits over the unemployment spell[4]. This paper therefore constitutes a bridge between moral hazard and equilibrium unemployment.

Landais, Michaillat, and Saez (2018) extend this logic by showing that the optimal design of unemployment insurance is dependent on labor market tightness. Adding the previously exposed counter

[3] Landais et al. (2015)(13) additionally propose a search and matching framework that models spillover effects, affecting labor market tightness (ratio between job seekers and vacancies). They find that firms can adjust the number of vacancies they create, and spillovers can also appear through crowding effects. However, it assumes that firms do not respond to changes in UI benefit levels through their vacancy creation channels

[4] In opposition, Cahuc and Lehmann (2000)(4) argue that declining benefits may significantly reduce the welfare for the long term unemployed workers, while Davidson and Woodbury (1997)(7) supported unlimited entitlements duration

Table 1: Recapitulative table of search incentives and job finding behaviors highlighted in the selected literature

Paper	**Model design**	**Main innovative feature**	**Parallel with our model**
Mortensen (1977)	Partial equilibrium job search model with endogenous search effort.	Provides the first micro founded framework linking unemployment insurance to search effort and unemployment duration. Is considered as a reference model in modeling unemployment insurance designs.	Justifies modeling endogenous search effort s_j, which determines job finding rates, with cost: $\frac{1}{2}c_0 s_j^2$.
Baily (1978)	Welfare based analysis of optimal UI framework balancing insurance and incentives.	Introduces the insurance-efficiency trade-off in UI design.	Provides the normative benchmark against which we evaluate policy changes, notably the trade-off between consumption smoothing and search distortions.
Chetty (2008)	Sufficient statistics approach using reduced form elasticities.	Assumes that optimal UI depends on two observables: consumption drop from unemployment and duration elasticity; distinguishing liquidity from moral hazard effects.	Guides the interpretation of behavioral responses in our model: changes in unemployment duration (2023 reform) incurs both incentive and liquidity effects.
Marinescu & Skandalis (2019)	Empirical search model with finite benefit duration and eligibility conditions, calibrated on French system & uses French administrative data.	Introduces within spell dynamics of search behavior and documents strong timing effects of UI duration, notably through a pre exhaustion surge.	Provides empirical support for a spike in search effort s_I as benefit exhaustion approaches, validating models with time varying search intensity and the role of duration λ in policy modulation.
Le Barbanchon, Rathelot & Roulet (2017)	Empirical analysis exploiting variation in benefit duration to study reservation wages and job-finding behavior.	Shows that reservation wages respond weakly to UI duration, challenging models relying on strong wage selectivity effects.	Justifies focusing on search effort rather than reservation wage as an adjustment mechanism, allowing for simplified wage modeling to evaluate the reform.
Landais, Michaillat & Saez (2018)	General equilibrium search and matching model with labor market tightness and welfare analysis over the business cycle.	Extends the classical Baily-Chetty formula by incorporating equilibrium tightness and the gap between micro and macro elasticities.	Motivates state dependent policy, and characterize optimal UI design as counter cyclical (through the macro correction term) including of labor market tightness and vacancy creation in the equilibrium evaluations.

cyclical correction term to the classical Baily-Chetty formula. In matching models, tightness is generally not efficient, so the optimal generosity of unemployment insurance cannot be derived from micro search responses alone. Our model shall hence allow for interactions between vacancy creation and market tightness, rather than solely focusing on job finding rates and individual outcomes. In line, Brébion, Briole, and Khoury (2025)(2) have studied French unemployment insurance eligibility rules and show that firms respond *ex ante* to those rules when choosing contracts duration. They demonstrate that unemployment insurance affects firms before matches are formed. Firms internalize workers preferences for contracts that satisfy eligibility thresholds, which shifts the distribution of posted jobs and generates spillover effects on workers already eligible for benefits. Rioux and Ortega (2010)(23) detail another useful mechanism through the notion of re-entitlement effects. When introducing non automatic eligibility, taking a job conditions future access to unemployment insurance, which alters the value of employment and matches through the prospect of future entitlements. This dimension is valuable when modeling the French scheme, where eligibility rules are a major policy lever. This paper shades lights on the fact that non automatic eligibility may attenuate the previously described mechanism, as workers would be likely to accept lower wages and would simultaneously soiling unemployment increase. Eligibility design is found to have a significant impact on equilibrium unemployment analysis and is implemented in our model through the three states condition.

From this literature review, we aim at constructing an appropriate framework that models to the best the French unemployment insurance system, allowing to seize dynamics from workers surplus and firms vacancy-posting decisions, analyzing consequences for matching efficiency and equilibrium unemployment.

Table 2: Recapitulative table of wage setting, vacancy creation and equilibrium unemployment publications

Paper	Model design	Main innovative feature	Parallel with our model
Fredriksson & Holmlund (1998)	Steady state general equilibrium search and matching model with Nash bargaining, free entry and endogenous search effort (relatively close to our model). UI affects simultaneously the value of unemployment (outside option) and the negotiated wage.	First paper to jointly models the search effort and wage bargaining channels of UI in a single equilibrium framework. Shows that the wage channel is quantitatively important and amplifies the standard moral hazard effect.	Motivates our model architecture with wage bargaining and vacancy creation.
Rioux & Ortega (2010)	Search model with non automatic endogenous eligibility and re-entitlement effects that together link employment to future UI access.	Introduces re-entitlement effects as a structural feature of UI design. Workers may accept lower wages to preserve future entitlement (forward looking incentive), affecting wage bargaining, job acceptance and finally equilibrium unemployment.	Models eligibility thresholds: a fundamental component of the French UI scheme, as well as the three-state structure (employed, eligible insured and non elegible workers) to support forward-looking behaviors embedded in V_E, V_I, V_U. Shows that eligibility design materially affects equilibrium wages and unemployment.
Brébion, Briole & Khoury (2025)	Empirical analysis of French administrative data exploiting eligibility thresholds to study firm ex ante firm behaviors to UI benefit rules.	Shows that firms adjust contract durations (ex ante), or prior to match formation, in response to UI eligibility rules as they internalize workers preferences. Hence affecting the distribution of vacancies and generating spillover effects on already-eligible workers.	Models firm responses in the vacancy creation equation and provides direct French empirical evidence that eligibility design shapes vacancy composition as well as worker behavior. Significant contribution for the analysis for the 2023 reform that changes the eligibility thresholds, and may affect the contracts designs offered by firms.

6 Transmission Channels of the Reform: A Steady-State Equilibrium Analysis

To assess whether the 2023 reform can achieve its three stated objectives — full employment, stronger work incentives, and reduced recruitment tensions — we develop a search-and-matching model with time-limited unemployment benefits and endogenous search effort, building on the Mortensen–Pissarides framework introduced above.

The model isolates two transmission channels:

- *Bargaining channel*: shorter entitlement weakens workers' outside options, moderating wages and stimulating vacancy creation.
- *Incentive channel*: workers anticipating benefit exhaustion intensify their search before entitlement expires.

Together, these channels determine equilibrium tightness θ, wages w, and search efforts s_I and s_U. All subsequent analysis compares the steady-state equilibria under the pre-reform and reformed benefit rules.

6.1 Environment

We consider a continuous-time search-and-matching economy populated by a unit mass of workers and a large number of firms. Workers can be employed, unemployed while receiving unemployment insurance, or unemployed without benefits. Unemployed workers in state $j \in \{I, U\}$ choose effort $s_j \in [0, 1]$, which scales their individual job-finding rate at a convex flow cost. Table 3 summarizes the labor market states, payoffs, and transition rates that characterize the environment.

States	E: employed $\quad I$: unemployed with benefits $\quad U$: unemployed without benefits
Payoffs	w in E, $\quad b_I$ in I, $\quad b_U$ in U
Search effort	$s_j \in [0,1]$ chosen by workers in state $j \in \{I, U\}$; cost $c(s_j) = \frac{1}{2} c_0 s_j^2$
Transitions	$E \xrightarrow{\delta} I, \quad I \xrightarrow{\lambda} U, \quad I, U \xrightarrow{s_j f(\theta)} E$
Matching Function	$m(u_e, v) = m_0\, u_e^{\eta} v^{1-\eta}, \quad u_e = s_I u_I + s_U u_U$
Output	Each match produces y
Tightness	$\theta = v/u_e$, where $u_e = s_I u_I + s_U u_U$ is the effective search pool
Job finding	$f(\theta) = m_0\, \theta^{1-\eta}$
Vacancy filling	$q(\theta) = m_0\, \theta^{-\eta}$

Table 3: Environment and labor market structure

The dynamics of the labor market are described by the laws of motion for employment, insured unemployment, and uninsured unemployment, reported in Table 4.

State	Law of motion	Economic interpretation
Employment (e)	$\dfrac{de}{dt} = s_I f(\theta)\, u_I + s_U f(\theta)\, u_U - \delta e$	Employment increases when unemployed workers find jobs (at effort-scaled rates) and decreases due to job destruction.
Insured unemployment (u_I)	$\dfrac{du_I}{dt} = \delta e - \big(\lambda + s_I f(\theta)\big) u_I$	Workers enter insured unemployment after job destruction and exit either by finding a job or exhausting benefits.
Uninsured unemployment (u_U)	$\dfrac{du_U}{dt} = \lambda u_I - s_U f(\theta)\, u_U$	Workers enter uninsured unemployment when benefits expire and exit when they find a job.

Table 4: Laws of motion of the labor market states

6.2 Modeling Unemployment Insurance

The model captures the key institutional feature of unemployment insurance: benefits are available for a limited duration. Unemployed workers initially receive unemployment insurance benefits and become uninsured once eligibility expires. Table 5 summarizes the benefit structure.

State	Flow income
Employment (E)	wage w
Insured unemployment (I)	benefit b_I
Uninsured unemployment (U)	income support b_U, with $b_U \leq b_I$

Table 5: Unemployment insurance benefit schedule

Benefit entitlement expires at hazard rate λ. Insured unemployed workers therefore transition from I to U when eligibility is exhausted. Labor market transitions are

$$E \xrightarrow{\delta} I, \qquad I \xrightarrow{\lambda} U, \qquad I \xrightarrow{s_I f(\theta)} E, \qquad U \xrightarrow{s_U f(\theta)} E.$$

Relevance for Evaluating the 2023 Reform

In our model, unemployed workers initially enter the insured state I and receive benefits b_I. Eligibility expires at rate λ, after which workers transition to uninsured unemployment U and receive a lower income level b_U. This reduced-form representation abstracts from the detailed eligibility rules of the French system (such as contribution requirements, age-dependent durations, or declining replacement rates). Instead, the parameter λ summarizes the expected duration of benefit entitlement. Changes in unemployment insurance duration — such as the modulation introduced by the 2023 reform — can therefore be interpreted in the model as changes in λ. Crucially, since λ also governs the speed at which workers exhaust benefits, a higher λ creates stronger incentives to search intensively while still insured, which is captured endogenously through the effort choice s_I.

6.3 Workers

Workers evaluate the expected lifetime value of each labor market state, taking into account wages, unemployment benefits, search costs, and the possibility of benefit exhaustion. In states I and U, each worker optimally chooses effort s_j to maximize their value function.

State	**Composition of the value**	**Bellman equation**
Employment (E)	Wage income w plus the risk of job destruction, which moves the worker to insured unemployment.	$rV_E = w + \delta(V_I - V_E)$
Insured unemployment (I)	Benefit income b_I, net of search cost $c(s_I)$, plus the possibility of finding a job at rate $s_I f(\theta)$ and the risk of benefit expiration.	$rV_I = b_I - \frac{1}{2}c_0 s_I^2 + s_I f(\theta)(V_E - V_I) + \lambda(V_U - V_I)$
Uninsured unemployment (U)	Lower income b_U, net of search cost $c(s_U)$, plus the possibility of finding a job at rate $s_U f(\theta)$.	$rV_U = b_U - \frac{1}{2}c_0 s_U^2 + s_U f(\theta)(V_E - V_U)$

Table 6: Worker value functions

6.4 Firms

Firms create jobs by posting vacancies and hiring workers through the matching process. Let J denote the value of a filled job and V_v the value of an open vacancy.

Object	Economic interpretation	Equation
Filled job (J)	Value of employing a worker. The firm receives output y, pays wage w, and faces the risk of job destruction.	$rJ = y - w + \delta(V_v - J)$
Vacancy (V_v)	Value of an unfilled vacancy. Firms pay a flow posting cost γ while waiting to meet a worker.	$rV_v = -\gamma + q(\theta)(J - V_v)$
Free entry	Firms post vacancies until the value of a vacancy is driven to zero.	$V_v = 0$

Table 7: Firm value functions and vacancy creation

Under free entry, firms create vacancies until the expected profit from posting an additional vacancy is driven to zero. This condition implies that the expected cost of hiring equals the expected surplus generated by a job match. The resulting job-creation condition is

$$\frac{\gamma}{q(\theta)} = \frac{y - w}{r + \delta}. \tag{1}$$

The derivation of equation (1) is provided in Appendix A.

6.5 Wage Determination

Wages are determined through Nash bargaining between workers and firms. When a match is formed, the worker and the firm split the surplus generated by the employment relationship. The worker's outside option corresponds to the value of insured unemployment V_I, since workers who separate from employment enter the insured state. Table 8 summarizes the components of the bargaining problem.

Object	Economic interpretation	Expression
Worker surplus	Gain for the worker from accepting employment rather than remaining in insured unemployment.	$V_E - V_I$
Firm surplus	Gain for the firm from employing a worker rather than keeping a vacancy open.	$J - V_v$
Bargaining rule	The worker receives share β of the match surplus and the firm receives $1 - \beta$.	Nash bargaining

Table 8: Wage bargaining

Applying Nash bargaining and substituting the optimal effort rules yields the following wage equation:

$$w = b_I - \tfrac{1}{2}c_0 s_I^2 + \big(r + \delta + s_I f(\theta) + \lambda\big)\frac{\beta}{1-\beta}\frac{\gamma}{q(\theta)} - \lambda\,\frac{c_0}{f(\theta)}\,s_U. \tag{2}$$

This expression generalises the standard Nash bargaining wage. The term $\frac{1}{2}c_0 s_I^2$ nets out the search cost borne by the worker in the outside option state. The term involving s_U captures the value of transitioning to uninsured unemployment: if uninsured workers search intensively (high s_U), the loss

from exhaustion is partly mitigated, which moderates wages. The derivation of equation (2) is provided in Appendix B.

Why do we use the Nash equilibrium?

The Nash bargaining solution (9) has become the standard framework for wage determination in the labor-market literature with search and matching frictions — not because of any strong empirical or theoretical justification, but largely for practical reasons: it is analytically tractable, has a clean game-theoretic foundation, and has simply become the benchmark against which other approaches are measured. That said, the solution has faced both theoretical and empirical criticism, and its dominance reflects convention as much as genuine superiority.

6.6 Equilibrium System

The equilibrium of the model is determined by four equations in four unknowns (θ, w, s_I, s_U).

$$\begin{cases} \dfrac{\gamma}{q(\theta)} = \dfrac{y-w}{r+\delta} \\[2ex] w = b_I - \frac{1}{2}c_0 s_I^2 + \big(r+\delta+s_I f(\theta)+\lambda\big)\dfrac{\beta}{1-\beta}\dfrac{\gamma}{q(\theta)} - \lambda\,\dfrac{c_0}{f(\theta)}\,s_U \\[2ex] s_I = \dfrac{f(\theta)}{c_0}\,\dfrac{\beta}{1-\beta}\,\dfrac{\gamma}{q(\theta)} \\[2ex] \dfrac{1}{2}c_0 s_U^2 + \dfrac{c_0(r+\lambda)}{f(\theta)}\,s_U + \left[(b_U-b_I)-(r+\lambda)\dfrac{\beta}{1-\beta}\dfrac{\gamma}{q(\theta)} - \dfrac{f(\theta)^2}{2c_0}\left(\dfrac{\beta}{1-\beta}\dfrac{\gamma}{q(\theta)}\right)^2\right] = 0 \end{cases} \tag{3}$$

Equation (i) is the job-creation condition. Equation (ii) is the wage equation. Equation (iii) is the optimal effort rule for insured workers: s_I rises with tightness and with the match surplus, so a reform that raises λ and lowers the worker's outside option V_I directly increases s_I. Equation (iv) is the analogous quadratic condition for uninsured workers.

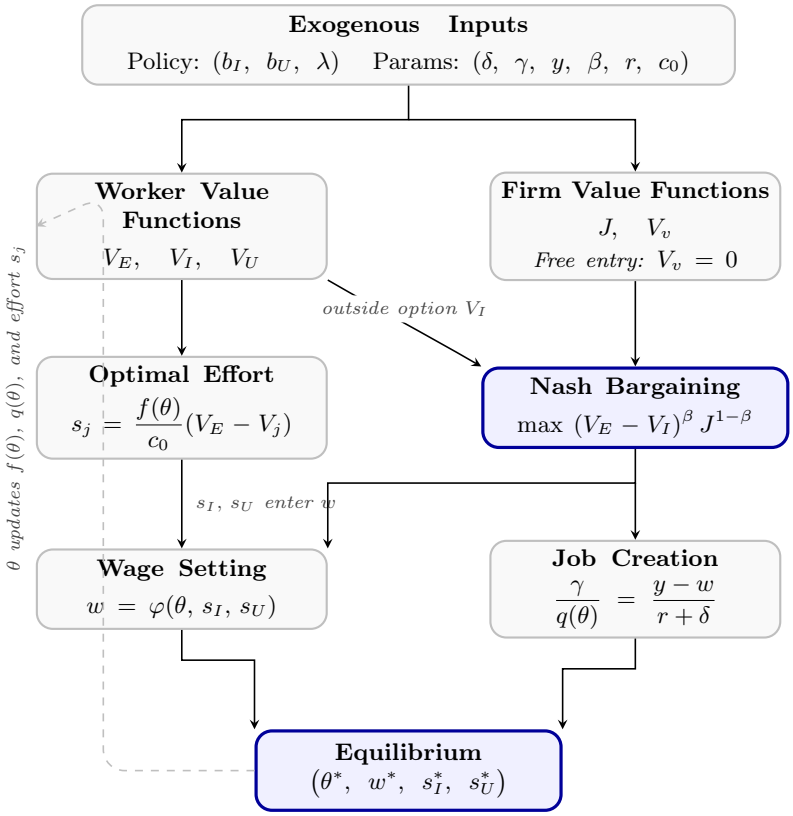


Figure 3: Model architecture: from exogenous inputs to equilibrium, with endogenous effort. The dashed arrow captures the fixed-point nature of the equilibrium.

Reading the figure. *Exogenous inputs — policy parameters* (b_I, b_U, λ) *and structural parameters* $(\delta, \gamma, y, \beta, r, c_0)$ *— feed simultaneously into worker and firm value functions. On the left branch, worker value functions determine optimal search effort* s_j *through the first-order condition of the worker's problem. On the right branch, worker and firm surpluses jointly enter Nash bargaining. The bargaining outcome feeds into both the wage-setting condition (WS) and the job-creation condition (JC), which together pin down the equilibrium quartet* $(\theta^*, w^*, s_I^*, s_U^*)$*. The dashed arrow captures the fixed-point nature of the equilibrium: market tightness* θ *affects the job-finding rate* $f(\theta)$*, the vacancy-filling rate* $q(\theta)$*, and therefore optimal effort* s_j*, so the four equations must be solved simultaneously.*

7 Calibration and Reform Simulation

7.1 Calibration

The model contains parameters that cannot be directly observed in the data — the vacancy posting cost γ, the matching efficiency m_0, and the income levels b_I and b_U. While the search-and-matching literature provides estimates for these parameters, they come from different countries, time periods, or institutional contexts, and cannot be directly applied to the French labor market in H2 2022. They serve as *orders of magnitude* to guide the search, not as targets. Calibration assigns values to these unknowns by finding the parameter vector that makes the model reproduce key features of the French

labor market in H2 2022.

The procedure follows two steps, illustrated in Figure 4.

Step 1 — Parameters fixed from the literature. Four parameters are assigned *a priori* and held fixed throughout. They never enter the optimization loop.

Parameter	Symbol	Value	Interpretation	Source
Quarterly interest rate	r	0.0125	5% annual ÷ 4	Standard
Bargaining power	β	0.500	Hosios condition	Hosios (5)
Match output	y	1.000	Normalization	—
Matching elasticity	η	0.500	Cobb-Douglas matching	Petrongolo and Pissarides (10)

Step 2 — Parameters estimated by the algorithm. The six remaining parameters (δ, λ, γ, m_0, b_I, b_U) are jointly estimated by minimizing the weighted distance between model-implied moments and their empirical counterparts:

$$\mathcal{L}(z) = \sum_k \omega_k \left(\frac{m_k(z) - \bar{m}_k}{\bar{m}_k} \right)^2$$

The algorithm starts from an initial guess z_0 grounded in the data and the literature: δ and λ are initialized at their empirically observed values, while γ, m_0, b_I, b_U are initialized from values reported in the search-and-matching literature. These are *orders of magnitude*, not constraints — the algorithm updates all six parameters freely until $\mathcal{L}$ is minimized.

For each candidate vector z, the equilibrium system (3) is solved numerically for (θ^*, w^*), and moments are computed from the resulting steady state.

Parameter	Symbol	Initial guess	Source
Job destruction rate	δ	0.014	Hairault et al. (4), Table 2
Benefit expiration hazard	λ	0.1875	Unédic (2022), 16-month avg.
Insured UI benefit	b_I	0.55	OECD replacement rate
Uninsured income	b_U	0.15	RSA + transfers, CAF (2022)
Vacancy posting cost	γ	0.40	Hagedorn and Manovskii (3)
Matching efficiency	m_0	0.60	Petrongolo and Pissarides (10)

Table 9: Initial guesses for Step 2. All six parameters are updated freely by the algorithm — these values only serve as starting points.

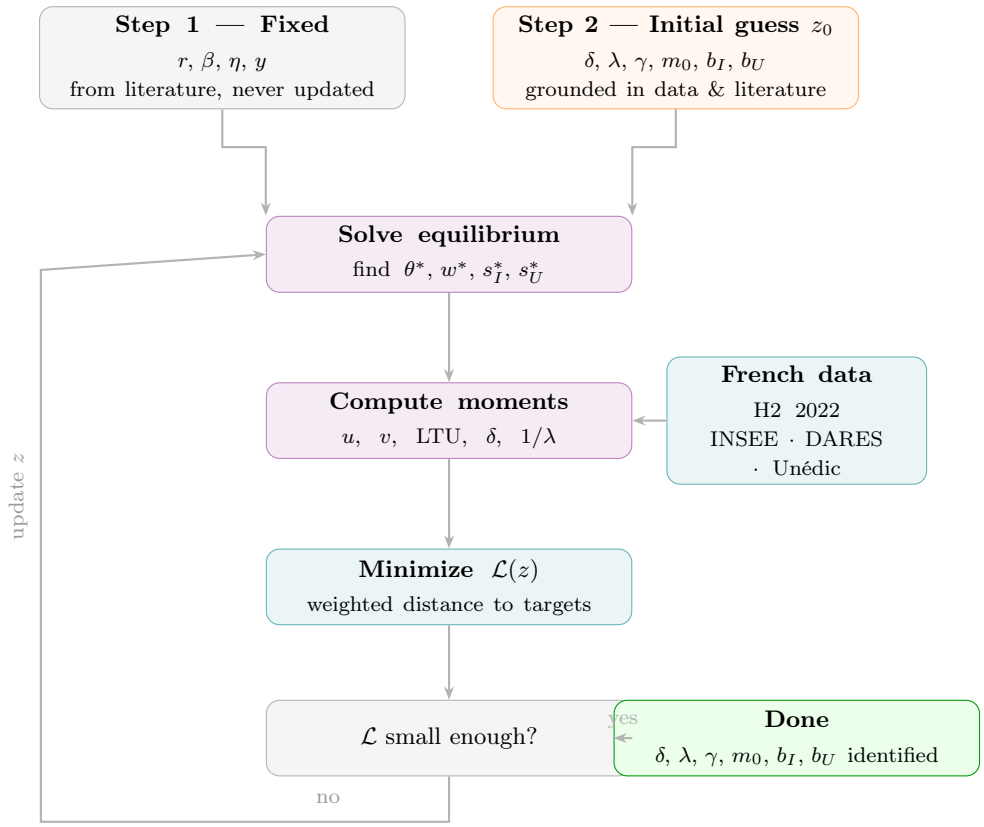


Figure 4: Calibration procedure. Step 1 parameters are fixed constants. Step 2 iterates over $z = (\delta, \lambda, \gamma, m_0, b_I, b_U)$, starting from data- and literature-grounded initial values, until $\mathcal{L}(z)$ is minimized.

Table 10 reports the model fit. The calibration matches the vacancy rate and UI duration closely. The total unemployment rate is modestly underestimated (−0.70 pp) while long-term unemployment is slightly overestimated (+0.72 pp)

Moment	Target	Model	Gap (pp)	Source
Total unemployment rate	7.30%	6.60%	−0.70	INSEE BIT, avg. T3–T4 2022
Vacancy rate	2.50%	2.41%	−0.09	DARES STMT, avg. T3–T4 2022
Long-term unemployment (≥6m)	3.50%	4.22%	+0.72	INSEE BIT, avg. T3–T4 2022
UI duration (quarters)	5.33	5.43	+0.10	Unédic, avg. indemnification

Table 10: Model fit: targets vs. calibrated moments (H2 2022 baseline).

7.2 Reform Simulation: Steady-State Comparative Statics

The reform is modeled as a 25% reduction in expected benefit duration. In the model, this is captured by an increase in the benefit expiration hazard:

$$\lambda_{\text{reform}} = \frac{\lambda_{\text{base}}}{0.75} = 0.2512, \qquad D_{\text{reform}} = \frac{1}{\lambda_{\text{reform}}} \approx 3.98 \text{ quarters.}$$

All other parameters are kept at their calibrated values. The effect of the reform is identified through a comparison of the steady-state equilibria associated with λ_{base} and λ_{reform}.

Interpreting the simulation: steady-state comparison, not a forecast

The simulation does **not** produce a forecast for any specific year. It compares two permanent steady-state equilibria: the *baseline*, calibrated to match H2 2022 labor market conditions, and a *counterfactual* in which benefit duration is permanently reduced by 25%, all else equal.

This approach answers the question: *if the French labor market had always operated under the reformed rules, what would the long-run equilibrium look like?* It identifies the **direction and magnitude of structural effects** — on wages, tightness, job creation, search effort, and unemployment composition — that the reform induces through its equilibrium mechanisms.

Two limitations follow directly. First, the model abstracts from **transition dynamics**: in reality, adjustment from one steady-state to another takes several quarters, and short-run effects may differ from long-run ones. Second, all non-policy parameters are held fixed, so the exercise isolates the effect of a change in λ **within the model environment** and does not capture interactions with other contemporaneous policy or macroeconomic changes.

7.2.1 Results

Aggregate unemployment and the dual mechanism. The steady-state comparison suggests that, within the model, a shorter expected duration of unemployment benefits reduces aggregate unemployment from 7.36% to 6.86% (Table 11). This improvement is driven by two complementary mechanisms:

- a 2.9% increase in the market job-finding rate $f(\theta)$, reflecting higher market tightness as θ rises by 6.0%;
- a 6.0% increase in the search effort of insured workers, s_I, reflecting stronger incentives to find employment before benefit exhaustion.

	Baseline	**Reform**	**Change**
Policy lever			
Benefit expiration hazard λ	0.1884	0.2512	+33.3%
Expected UI duration $D = 1/\lambda$	5.31 qtrs	3.98 qtrs	−25.0%
Equilibrium outcomes			
Market tightness θ	0.3388	0.3590	+6.0%
Wage w	0.8801	0.8766	−0.4%
Market job-finding rate $f(\theta)$	0.2388	0.2458	+2.9%
Vacancy-filling rate $q(\theta)$	0.7047	0.6846	−2.9%
Search effort			
Insured effort s_I	0.5958	0.6313	+6.0%
Uninsured effort s_U	0.8872	0.8869	−0.0%
Unemployment decomposition			
Insured unemployment i	3.89%	3.19%	−0.71pp
Uninsured unemployment u	3.46%	3.67%	+0.21pp
Total unemployment $i + u$	**7.36%**	**6.86%**	**−0.50 pp**
Employment e	92.64%	93.14%	+0.50 pp
Matching pool			
Effective search pool u_e	0.0539	0.0527	−2.3%
Vacancy stock v	0.0183	0.0189	+3.5%

Table 11: Steady-state comparison: baseline (H2 2022) vs. reform counterfactual. Change column reports percentage change, except unemployment rates which are in percentage points.

Wage moderation and vacancy dynamics. At the same time, the reform generates moderate wage moderation: equilibrium wages fall by 0.4%, which raises firms' expected surplus and supports vacancy posting. The vacancy stock rises modestly in absolute terms (from 0.0183 to 0.0189, +3.5%), driven by improved expected profitability. However, the increase in market tightness also reduces the vacancy-filling rate by 2.9%, implying that each individual vacancy takes longer to fill even as the overall vacancy stock expands slightly.

Asymmetry of the search effort response. A key result concerns the asymmetry of the effort response. The reform raises s_I by 6.0%, as insured workers anticipate the shorter entitlement horizon and intensify their search before exhaustion. By contrast, s_U is virtually unchanged (−0.0%): uninsured workers already search near maximum intensity at baseline, so further compressing their outside option yields no additional margin.

Composition of unemployment. A central finding is the change in the composition of unemployment. Insured unemployment declines by 0.71pp, from 3.89% to 3.19%, while uninsured unemployment rises by 0.21pp, from 3.46% to 3.67%. The net effect is a −0.50pp reduction in total unemployment. Hence, the reform improves aggregate labor market performance, but does so partly through a reallocation from insured to uninsured unemployment. The model thus highlights a trade-off between lower total unemployment and weaker income protection among the unemployed.

Firm-side response and vacancy posting. A result that warrants attention concerns the firm-side implications of the reform. Although market tightness rises by 6.0%, this reflects a contraction of the effective search pool u_e rather than an explosion of vacancy posting. The effective search pool falls by 2.3% (from 0.0539 to 0.0527), as fewer workers are unemployed overall, while the vacancy stock rises modestly by 3.5% (from 0.0183 to 0.0189). Tightness therefore rises because the pool of effective searchers contracts faster than vacancies expand. The employment gain is driven primarily by the worker side — stronger search effort and a thinner unemployment pool — rather than by an active firm-side response. Figure 5 summarizes this transmission.

Implications for recruitment. This has direct implications for recruitment. The vacancy-filling rate $q(\theta)$ falls by 2.9%, meaning each individual vacancy takes longer to fill under the reform. Firms thus face a squeeze: the vacancy stock rises slightly in absolute terms, but each vacancy becomes harder to fill as the effective search pool contracts. In aggregate this is a second-order effect — the unemployment rate still falls by 0.50 pp — but it is not negligible at the margin. In sectors already reporting structural recruitment difficulties at the H2 2022 calibration date, a further compression of $q(\theta)$ may constrain the hiring response, particularly where vacancy duration is already long and replacement costs are high. The model cannot disaggregate by sector, so this remains a quantitative caveat rather than a model-identified result. What the model does establish is that the reform's employment gains are purchased partly at the cost of modestly worse recruiting conditions for firms.

7.3 Policy Implications

Four implications follow from the results. First, **the -0.50 pp decline in unemployment is directionally consistent with the full employment objective but quantitatively modest**: reaching a 5% unemployment rate from the 7.36% baseline would require a reduction more than four times larger. Duration cuts alone cannot substitute for broader supply-side reforms targeting δ and m_0.

Second, **the incentive channel is the central amplification mechanism**. The reform raises s_I by 6.0% through its direct effect on workers' anticipation of exhaustion, while the market job-finding rate rises by only 2.9%. The individual job-finding rate of insured workers, $s_I f(\theta)$, therefore rises by approximately 9.1% — a combination of stronger effort and a tighter market. Without the incentive channel — as in a model with exogenous effort — the reform would reduce unemployment by only -0.24 pp rather than -0.50 pp. The behavioral response of workers thus accounts for more than half of the total employment effect.

Third, the **composition shift is the central welfare trade-off**. The reform acts as a *benefit exhaustion accelerator*: insured unemployment falls by 0.71pp while uninsured unemployment rises

by 0.21pp, because the job-finding rate does not increase fast enough to absorb all workers before exhaustion. Aggregate unemployment improves, but at the cost of weaker income protection. The calibrated parameters reveal the starkness of this trade-off: income drops from $b_I = 0.800$ to $b_U = 0.418$ at exhaustion — a 48% fall — for workers who do not find employment in time.

Fourth, **the reform offers no relief to firms facing structural recruitment difficulties**. Although the vacancy stock rises modestly (+3.5%), the vacancy-filling rate falls by 2.9% as the effective search pool contracts. Sectors already operating under tight recruiting conditions at the H2 2022 baseline — where vacancy duration is long and replacement costs are high — should not expect the reform to ease their hiring constraints. If anything, the compression of $q(\theta)$ marginally worsens their position. This has implications for how the reform interacts with sectoral labor shortages, a dimension the aggregate model cannot capture but that empirical evaluation of the reform should monitor.

Fifth, H2 2022 is likely a *favorable case* for the reform. A tight labor market amplifies both the incentive channel and the bargaining channel. Under slack conditions, the same increase in λ would generate weaker job-creation effects, a smaller rise in s_I, and a larger accumulation of uninsured unemployed.

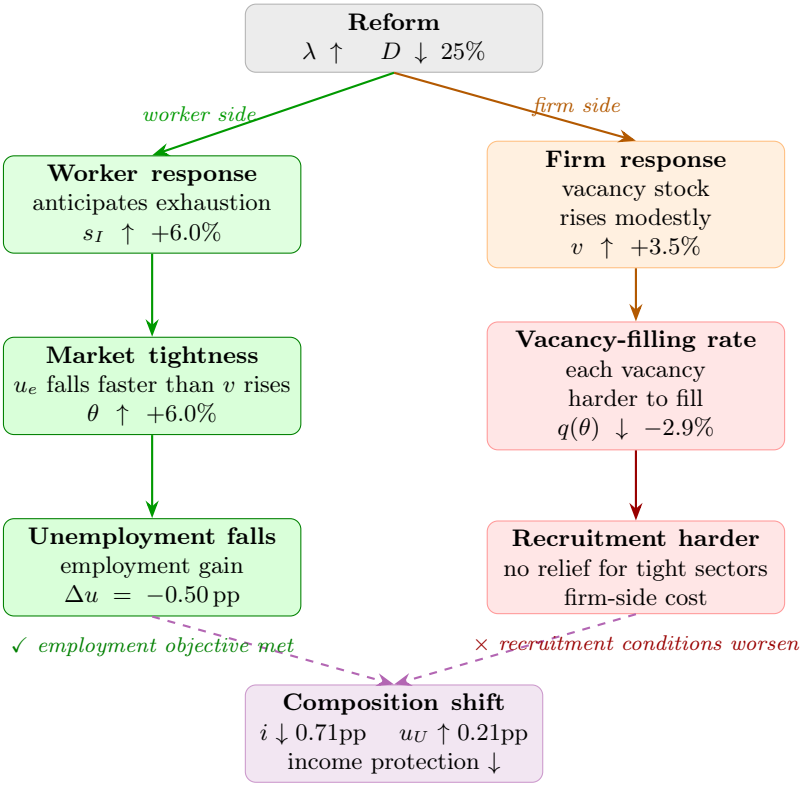


Figure 5: Firm-side vs. worker-side transmission of the reform. Market tightness rises because the effective search pool u_e contracts faster than vacancies expand. The vacancy-filling rate declines, worsening recruiting conditions despite the modest increase in vacancy posting. Both channels feed into a common welfare trade-off (purple).

The steady-state framework, by construction, abstracts from the dynamics that characterize real labor markets. It identifies the long-run equilibrium effects of the reform under a fixed macroeconomic environment, but remains silent on how those effects evolve as conditions change. This is a fundamental limitation for evaluating a reform whose central design principle is precisely that policy should respond to the state of the economy. Steady-state analysis cannot speak to transition dynamics, cannot assess whether institutional triggers activate at the right moment, and cannot identify which workers bear the cost of adjustment between regimes. To address these limitations, we extend the model to incorporate aggregate productivity shocks through a Markov chain over discrete states, following Mortensen and Pissarides (8). Rather than a single fixed equilibrium, the economy now transitions between a good state, a normal state, and a bad state, each associated with its own tightness, destruction threshold, and unemployment rate. This extension allows the reform to be evaluated not against a fixed baseline, but across the business cycle it was explicitly designed to govern.

8 Wage Rigidities: A Minimum Wage Extension

France operates one of the most binding minimum wage regimes among OECD economies. The *Salaire Minimum Interprofessionnel de Croissance* (SMIC) sets a statutory wage floor that applies across all sectors and is regularly indexed to inflation and productivity growth, covering a disproportionate share of young and low-skilled workers. It is therefore of interest to study an extension of the baseline model that incorporates wage rigidity.

8.1 Wage Rigidity and Equilibrium Structure

The environment, matching technology, and benefit structure remain exactly as specified in Section 6.1. The only modification concerns wage determination: rather than allowing workers and firms to bargain freely, we impose a wage floor w_0 below which the negotiated wage cannot fall.

Two regimes. If $w_0 < w^*$, the constraint is non-binding and the equilibrium of system (3) is unchanged. The interesting case arises when $w_0 \geq w^*$: bargaining can no longer deliver the unconstrained outcome, and the equilibrium wage is pinned at the floor, $w = w_0$, independently of market tightness θ.

Structural consequence. In the unconstrained model, θ and w are jointly determined through system (3): tightness affects wages through workers' outside options, and wages in turn affect vacancy creation. When $w = w_0$ is imposed exogenously, this simultaneity breaks down. The wage is no longer a variable to be solved for. The equilibrium becomes *sequential*: market tightness θ_0 is determined first from the job-creation condition alone, and search efforts (s_I, s_U) are determined subsequently.

This triangular structure contrasts with the fixed-point nature of system (3), where all four unknowns had to be solved simultaneously.

8.2 Equilibrium System

The equilibrium under the minimum wage is characterised by three equations in three unknowns (θ_0, s_I, s_U).

$$\begin{cases} \dfrac{\gamma}{q(\theta_0)} = \dfrac{y - w_0}{r + \delta} \\[2ex] (r + \delta + \lambda)\,\dfrac{c_0\, s_I}{f(\theta_0)} = (w_0 - b_I) - \tfrac{1}{2} c_0 s_I^2 + \lambda\,\dfrac{c_0\, s_U}{f(\theta_0)} \\[2ex] (r + \lambda)\,\dfrac{c_0 (s_U - s_I)}{f(\theta_0)} = (b_I - b_U) + \tfrac{1}{2} c_0 \left(s_I^2 - s_U^2\right) \end{cases} \tag{4}$$

Equation (i) is the job-creation condition, derived in Appendix A with w_0 replacing the Nash wage. Equation (ii) governs the effort of insured workers: it is obtained from the employment surplus expression without invoking the Nash sharing rule, which no longer holds when $w = w_0$. Equation (iii) governs the effort of uninsured workers: it relies solely on the Bellman equations and workers' first-order conditions, and is therefore valid regardless of how wages are set. The derivations of equations (ii) and (iii) are provided in Appendix C.

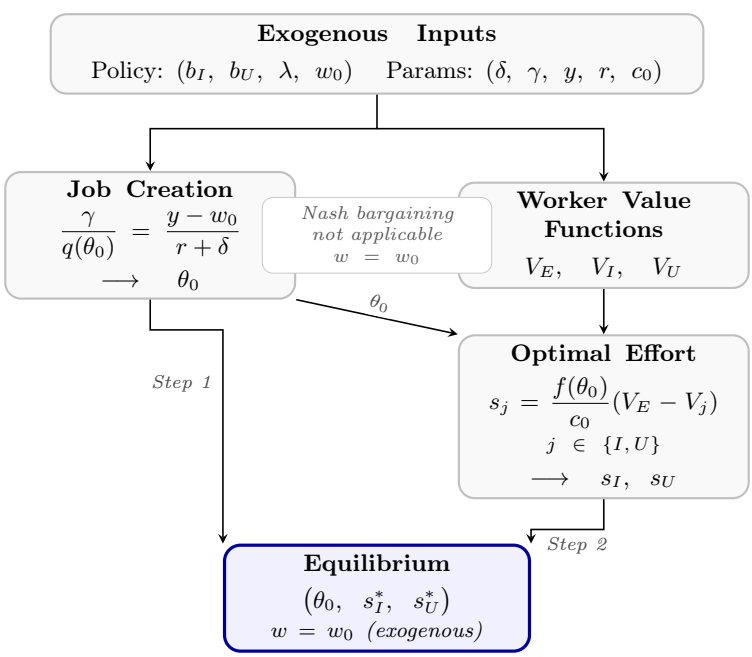


Figure 6: Sequential equilibrium structure under the minimum wage. In Step 1, market tightness θ_0 is pinned by the job-creation condition with $w = w_0$, independently of worker behaviour. In Step 2, worker value functions determine optimal search efforts s_I and s_U given θ_0. Nash bargaining plays no role: the wage is fixed exogenously, severing the feedback loop between tightness, wages, and outside options.

8.3 Equilibrium Effects

Market tightness. The effect on tightness is unambiguous. Since $w_0 \geq w^*$, the right-hand side of equation (i) is weakly smaller than under Nash bargaining. Since $q(\theta) = m_0\,\theta^{-\eta}$ is strictly decreasing, this implies:

$$\theta_0 \leq \theta^*.$$

The wage floor compresses firms' expected surplus from a match, discourages vacancy posting, and reduces market tightness.

Search effort. The effects on s_I and s_U are theoretically ambiguous. For insured workers, the first-order condition $c_0 s_I = f(\theta_0)\,\Delta_I$ is subject to two opposing forces: the fall in θ_0 reduces $f(\theta_0)$ and tends to decrease s_I, while the higher wage $w_0 > w^*$ raises V_E and therefore Δ_I, tending to increase s_I. The same tension applies to uninsured workers through equation (iii) of system (4). Neither effect can be signed without a numerical solution.

8.4 Comparison with the Baseline Model

The minimum wage extension modifies the baseline model along four distinct dimensions.

Surplus sharing and worker rent. Under Nash bargaining, the surplus was split exactly according to bargaining weights: the worker received βS and the firm $(1-\beta)S$. When $w_0 > w^*$, this rule no longer holds. The worker captures more than βS and the firm less than $(1-\beta)S$. The difference $w_0 - w^* > 0$ constitutes a per-period rent accruing to the worker, entirely at the expense of the firm's surplus.

Match viability and non-creation. In the baseline model, any match with $y > rV_I$ was viable and would be formed. Under the wage floor, the firm's participation constraint requires $J = (y - w_0)/(r+\delta) \geq 0$, i.e. $y \geq w_0$. Matches whose output falls below the floor are not profitable for the firm and are never created. The wage floor does not merely redistribute surplus — it destroys it for low-productivity matches.

Asymmetric job destruction. In the baseline model, both parties agree to separate simultaneously when the match surplus S reaches zero. Under the wage floor, a wedge opens between the two parties' continuation values. There exists a range of productivity levels where the total surplus remains positive — so the worker prefers to remain employed — but the firm's individual surplus J is zero or negative. In this range, the firm wishes to destroy the job before the worker does. This asymmetry is a qualitative departure from the bilateral efficiency of the Nash solution, where separation was always mutually agreed upon.

Neutralisation of the bargaining channel. A central transmission mechanism of the 2023 reform in the baseline model was the bargaining channel: a higher λ weakened workers' outside options, lowered the Nash wage, and stimulated vacancy creation. Under the wage floor, this channel is entirely shut down. Since $w = w_0$ is invariant to λ, changes in benefit duration no longer affect wages. The incentive channel — through which λ raises s_I — remains operative, but its effect on tightness is attenuated: firms do not respond to stronger worker search effort by posting more vacancies, since the wage they must pay is fixed regardless. "'

8.5 Reform Simulation: Steady-State Comparative Statics

The reform is modeled identically to Section 6: a 25% reduction in expected benefit duration, captured by an increase in the benefit expiration hazard:

$$\lambda_{\text{reform}} = \frac{\lambda_{\text{base}}}{0.75} = 0.2512, \qquad D_{\text{reform}} = \frac{1}{\lambda_{\text{reform}}} \approx 3.98 \text{ quarters.}$$

The minimum wage is set at $w_0 = 0.9$. This value exceeds the Nash bargaining wage of the baseline model ($w^* = 0.8801$, Table 11), so the constraint is binding, as required.

8.5.1 Results

Neutralisation of the bargaining channel. The central result of the simulation is the complete absence of any tightness response to the reform (Table 12). Market tightness θ_0 is unchanged at 0.3390 under both the baseline and the reform counterfactual. This is a direct consequence of the minimum wage: since $w = w_0$ is invariant to λ, the job-creation condition

$$\frac{\gamma}{q(\theta_0)} = \frac{y - w_0}{r + \delta}$$

is unaffected by the change in benefit duration. The bargaining channel — through which, in the baseline model, a higher λ lowered the Nash wage and stimulated vacancy creation — is entirely shut down. Correspondingly, $f(\theta)$ and $q(\theta)$ are unchanged.

	Baseline	**Reform**	**Change**
Policy lever			
Benefit expiration hazard λ	0.1884	0.2512	+33.3%
Expected UI duration $D = 1/\lambda$	5.31 qtrs	3.98 qtrs	−25.0%
Equilibrium outcomes			
Market tightness θ_0	0.3390	0.3390	+0.0%
Market job-finding rate $f(\theta)$	0.2097	0.2097	+0.0%
Vacancy-filling rate $q(\theta)$	0.6185	0.6185	+0.0%
Search effort			
Insured effort s_I	0.6785	0.7223	+6.5%
Uninsured effort s_U	1.0105	1.0082	−0.2%
Unemployment decomposition			
Insured unemployment i	3.89%	3.21%	−17.6%
Uninsured unemployment u	3.46%	3.81%	+0.35pp
Total unemployment $i + u$	**7.36%**	**7.02%**	**−0.33 pp**
Employment e	92.64%	92.98%	+0.33pp

Table 12: Steady-state comparison under the minimum wage: baseline vs. reform counterfactual ($w_0 = 0.9$). Change column reports percentage change, except unemployment rates which are in percentage points.

Incentive channel remains operative. Despite the neutralisation of the bargaining channel, the reform retains its effect on search incentives. The effort of insured workers rises by 6.5%, from $s_I = 0.6785$ to $s_I = 0.7223$, as workers anticipate the shorter entitlement horizon and intensify their search before exhaustion. Since $f(\theta)$ is unchanged, this translates directly into a higher individual job-finding rate for insured workers, $s_I f(\theta_0)$, which drives the reduction in insured unemployment.

By contrast, s_U is virtually unchanged (−0.2%) and remains above 1 ($s_U \approx 1.01$), indicating that uninsured workers already operate at the boundary of the effort constraint. This is a technical limit of the model: the theoretical assumption $s_j \in [0, 1]$ (because $s_j f(\theta)$ is a probability) is marginally violated at the calibrated parameter values, suggesting that the effort cost c_0 may be slightly underestimated for this regime. The result is nonetheless interpretable: uninsured workers search at maximum intensity regardless of the reform, so the reform has no margin to act on their behaviour.

Weaker employment effect. The reform reduces total unemployment from 7.36% to 7.02%, a decline of 0.33pp. This is meaningfully smaller than the −0.50pp obtained in Section 6, where both the incentive and bargaining channels were active. The attenuation is entirely attributable to the absence of the tightness response: in the baseline model, θ rose by 6.0%, amplifying the employment gain through higher vacancy creation. Here, θ does not move, and the entire employment effect rests on the incentive channel alone.

Composition of unemployment. The compositional pattern mirrors that of Section 6. Insured unemployment falls sharply (−17.6%, from 3.89% to 3.21%), while uninsured unemployment rises by

0.35pp (from 3.46% to 3.81%). The reform accelerates benefit exhaustion without generating sufficient new employment to absorb all exiting insured workers before their entitlement expires. The aggregate employment gain is thus accompanied by a deterioration in income protection for those who do not find work in time, with income dropping from $b_I = 0.765$ to $b_U = 0.123$ at exhaustion.

Comparison with the baseline model. Table 13 summarises the contrast between the two simulations. The minimum wage does not prevent the reform from reducing unemployment, but it halves the tightness response and reduces the total employment gain by one third. The key insight is that the SMIC severs the link between benefit duration and vacancy creation: firms do not respond to the reform because the wage they must pay is fixed regardless of workers' outside options.

	Baseline model (Section 6)	**Minimum wage (Section 8)**	**Difference**
$\Delta\theta$	+6.0%	0.0%	−6.0pp
Δs_I	+6.0%	+6.5%	+0.5pp
Δs_U	−0.8%	−0.2%	+0.6pp
Δ total unemployment	−0.50pp	−0.33pp	+0.17pp
Bargaining channel	Active	Shut down	—
Incentive channel	Active	Active	—

Table 13: Comparison of reform effects: baseline model vs. minimum wage extension.

9 The Reform over the Business Cycle: Steady-State Analysis with Endogenous Job Destruction

The previous search and match model assumed the natural unemployment rate responded to shocks and parameter changes only through job creation. However, empirical evidence suggests that both job creation and job destruction respond to exogenous shocks and that they do so differently depending on the aggregate state of the economy. To understand the 2023 reform in a more realistic framework, we build a model that introduces endogenous job destruction by allowing differentiation in job-specific productivity. In this setting, jobs with zero or negative surplus are destroyed, which occurs when jobs' productivity fall below a given threshold, called 'reservation productivity'. Contrarily to the previous model, this threshold is endogenously determined. In addition, we extend the model to a three-state macroeconomic environment, in which a common productivity component across jobs varies across aggregate states, thereby generating business cycle fluctuations in job creation and job destruction.

The model closely follows that presented in section 6, with three main distinctions :

- **No endogenous search effort** : While search frictions remain, as in standard search and match models, workers do not optimally choose search effort. This modeling choice is done for

the sake of simplicity and interpretability of the model mechanisms.

- **Endogenous job destruction**: While the baseline model assumes homogeneous job productivity normalized to one, we introduce heterogeneity in jobs' productivity through a job-specific productivity component. This generates a distribution of match-specific productivity levels and implies the existence of a reservation threshold below which jobs are not profitable and are endogenously destroyed.

- **Three-State Macroeconomic Environment**: To capture business cycle dynamics, we model the economy as a three-state Markov process (high, intermediate, and low productivity states). Aggregate shocks affect the common component of job productivity and are calibrated to match French quarterly productivity levels and transition probabilities across states. This framework makes both the reservation productivity threshold and market tightness state-dependent, thereby generating cyclical variations in job creation and job destruction.

The model presented below follows (8).

9.1 Environment

9.1.1 Jobs Differentiation and Endogenous Job Destruction

In the previous model, exogenous shocks destroyed unprofitable jobs at an arrival rate μ, forcing some jobs to transition from a productivity $y = 1$ to $y = 0$. To allow for endogenous job destruction, we define job productivity as $p_i + \sigma y$, where p_i represents a common productivity component shared across all jobs that varies with the macroeconomic state $i = boom, intermediate, recession$, and σy captures a job-specific productivity component, introducing job-specific productivity variation. y follows $G(y)$, and, without loss of generality, we can assume $y \sim U[y_{min}, \bar{y}]$, while σ captures the dispersion of idiosyncratic productivity.

Idiosyncratic productivity shocks arrive at rate μ, changing a job's productivity from y to y'. The productivity draw following a shock is independent of the previous productivity level and is irreversible. As a result, the value of a filled job depends on its current productivity level, denoted by $J(y)$. After a shock, firms decide either to continue producing at idiosyncratic productivity y' or closing the job. Matches whose productivity remains above a reservation productivity threshold survive ($J(y') \geq 0$), whereas those falling below it are endogenously destroyed because they generate negative surplus.

This job destruction rule satisfies the reservation property defined by $J(R) = 0$, where R denotes the reservation productivity. In addition, since wage is tied to match surplus through wage setting as in the previous model, the reservation rule also satisfies that the worker will want to quit the job at

this productivity level, given that the value of remaining employed will be equivalent to that of being unemployment insured. Table 14 summarizes job destruction / continuation decision.

At job creation, the firm has complete choice of job productivity. Firms' profit maximization trivially requires that all new jobs are created at the maximal productivity level $\bar{y}$.

Job Productivity y_i	**Firm Surplus** $J(y_i)$	**Worker Value** $V_e(y_i)$	**Action**
$y_i < R_i$	$J(y_i) < 0$	$V_e(y_i) < V_{I_i}$	Job destroyed / worker quits
$y_i = R_i$	$J(R) = 0$	$V_e(R_i) = V_{I_i}$	Job destroyed / worker quits
$y_i > R_i$	$J(y_i) > 0$	$V_e(y_i) > V_{I_i}$	Job maintained; worker stays

Table 14: Reservation Productivity Threshold

9.1.2 Macroeconomic States and State-Dependent Job Creation and Job Destruction

We introduce the possibility of aggregate productivity to change probabilistically. We model three aggregate states of the economy : boom, intermediate and recession, characterized by productivity levels $p_1 > p_2 > p_3$, respectively. Transitions between states or aggregate shocks are captured in a transition matrix Ω_{ij}. Workers and firms are forward-looking: decisions on vacancy posting, job acceptance and job destruction depend on both the current productivity state and expectations about future states. Anticipated booms or recessions shape match formation, wages, and unemployment, allowing the model to capture the propagation of macroeconomic shocks through the labor market. As a result, equilibrium job creation and job destruction, and consequent unemployment, are state dependent.

9.1.3 Unemployment Dynamics

Inflows to unemployment, determined by job destruction, are now endogenous to the model and state-dependent. As in the previous model, outflows from unemployment, determined by job creation, are endogenous as it depends on market tightness but are now state-dependent too. Benefit expiration, and the consequent transition from insured to uninsured unemployment, remains unchanged. This is done for simplicity of the exposition of the model. A model extension making unemployment benefits and social assistance tied to the macroeconomic states is, however, totally feasible, and would enrich the observed unemployment dynamics. Table 9.1.3 summarizes the unemployment dynamics.As in the previous model, equilibrium unemployment imply that inflows to unemployment equal outflows.

$$E \xrightarrow{\mu G(R_i)} I, \qquad I \xrightarrow{\lambda} U, \qquad I \xrightarrow{q(\theta_i)} E, \qquad U \xrightarrow{q(\theta_i)} E.$$

Concept	Expression	Logic
Unemployment flow	$\dot{u}_i = \mu G(R_i)(1-u_i) - \theta_i q(\theta_i) u_i$	Unemployment changes with inflows from job destruction and outflows from job finding.
Steady-state unemployment - Beveridge Curve	$u_i = \dfrac{\mu G(R_i)}{\mu G(R_i) + \theta_i q(\theta_i)}$	In steady state, inflows into unemployment equal outflows. The unemployment rate depends on separation vs. matching rates.
Insured unemployment flow	$\dot{u}_i^I = \mu G(R_i)(1-u_i) - (\theta_i q(\theta_i) + \lambda) u_i^I$	Insured unemployed enter employment through job destruction and leave either by finding a job or by exhausting benefits.
Steady state insured unemployment	$u_i^I = \dfrac{\mu G(R_i)(1-u_i)}{\theta_i q(\theta_i) + \lambda}$	Stock of unemployed receiving benefits.
Uninsured unemployment flow	$\dot{u}_i^U = \lambda u_i^I - \theta_i q(\theta_i) u_i^U$	Workers enter uninsured unemployment when benefits expire and exit when they find a job.
Steady State Uninsured unemployment	$u_i^U = \dfrac{\lambda}{\theta_i q(\theta_i)} u_i^I$	Stock of unemployed after benefit exhaustion. Implies a flows out of uninsured unemployment equal inflows.

Table 15: Labor market states: employment and unemployment dynamics and steady states.

9.1.4 Workers

Table 16 summarizes the worker's dynamic problem across employment and unemployment states. Each value function captures the expected discounted payoff in that state, accounting for immediate income, transitions across employment states, idiosyncratic shocks, aggregate macroeconomic transitions between states. Workers make forward-looking decisions by comparing the value of remaining in their current state to the alternatives, which determines their job acceptance behavior which gives their reservation wage.

State	Composition of the value	Bellman equation
Employement (V_{e_i})	Value of employment. The worker receives a wage and faces gains or losses from idiosyncratic productivity shocks at a rate (μ) and transitions between aggregate economic states at transition probabilities (Ω_{ij}), remaining employed only if the new state's value exceeds the value of insured unemployment.	$rV_{e_i}(y) = w_{e_i}(y) + \mu\left[\int \max\{V_{e_i}(y'), V_{I_i}\}\, dG(y') - V_{e_i}(y)\right] + \sum_{j\neq i}^{n} \Omega_{ij}\left[\max\{V_{e_j}(y), V_{I_i}\} - V_{e_i}(y)\right]$
Unemployed insured (V_{I_i})	Value of insured unemployment. The worker receives an unemployment benefit b_{I_i}, transitions to employment with probability $\theta_i q(\theta_i)$, faces the risk of losing insurance at a rate (λ) and transitions between aggregate economic states at (Ω_{ij}).	$rV_{I_i} = b_I + \theta_i q(\theta_i)\left(V_{e_i}(\bar{y}) - V_{I_i}\right) + \lambda\left(V_{U_i} - V_{I_i}\right) + \sum_{j\neq i}^{n} \Omega_{ij}\left[V_{I_j} - V_{I_i}\right]$
Unemployed uninsured (V_{U_i})	Value of uninsured unemployment. The worker receives social assistance b_U, transitions to employment with probability $\theta_i q(\theta_i)$ and between aggregate economic states at (Ω_{ij}).	$rV_{U_i} = b_{U_i} + \theta_i q(\theta_i)\left[V_{e_i}(\bar{y}) - V_{U_i}\right] + \sum_{j\neq i}^{n} \Omega_{ij}\left[V_{U_j} - V_{U_i}\right]$

Table 16: Worker value functions

9.1.5 Firms

Table 17 summarizes the firm's dynamic problem across filled jobs and vacancies. Each value function captures the expected discounted payoff for the firm, accounting for immediate net profits, idiosyncratic productivity shocks and aggregate macroeconomic transitions between states.The table also re-introduces the critical free entry condition that underpins firm's job creation decisions. As workers, firms make forward-looking decisions by comparing the value of continuing a filled job versus terminating it and by evaluating whether to post new vacancies, which determines **optimal vacancy posting and labor demand** in response to both idiosyncratic and aggregate shocks.

Object	Economic interpretation	Equation
Filled job (J)	Value of filled job. Firms obtain net profit (output $p_i + \sigma y$ minus wage $w_i(y)$) plus the expected gains from idiosyncratic productivity shocks (μ) and transitions between aggregate economic states (Ω_{ij}), subject to the firm's option to terminate the position if its value falls below the value of a vacancy.	$rJ_i(y) = p_i + \sigma y - w_i(y) + \mu\left[\int \max\{J_i(y'), V_i\}\, dG(y') - J_i(y)\right] + \sum_{j\neq i}^{n} \Omega_{ij}\left[\max\{J_j(y), V_j\} - J_i(y)\right]$
Vacancy (V_v)	Value of a vacancy. Vacancies cost $-\gamma$ plus the expected capital gains from successfully matching with a worker at rate $q(\theta)$ and from the transition of the vacancy's value across different aggregate economic states (Ω_{ij}).	$rV_{vi} = -\gamma + q(\theta)\left[J_i(\bar{y}) - V_i\right] + \sum_{j\neq i}^{n} \Omega_{ij}\left[V_{vj} - V_{vi}\right],$
Free entry	In equilibrium, firms post vacancies until the value of a vacancy is driven to zero.	$V_{vi} = 0$

Table 17: Firm value functions and vacancy creation

9.2 Steady State Equilibrium

9.2.1 Model Resolution

We solve the model by expressing equilibrium conditions in terms of the match surplus, which measures the value of a worker–firm match relative to their respective outside options. This surplus is the key object governing both the formation and destruction of matches. Using the surplus, we derive two central equilibrium conditions:

1. **Job creation condition**: obtained from the free-entry condition and the surplus sharing rule, determines firms' incentives to post vacancies and pins down labor market tightness, θ.

2. **Job destruction condition**: obtained from reservation property and surplus sharing rule, identifies reservation productivity R_i below which matches are no longer profitable and are endogenously terminated.

Together, these two conditions jointly determine equilibrium (R_i, θ_i). Given these equilibrium objects, the **unemployment rate** can then be recovered from the Beveridge curve, linking job finding and job destruction flows to aggregate labor market outcomes.

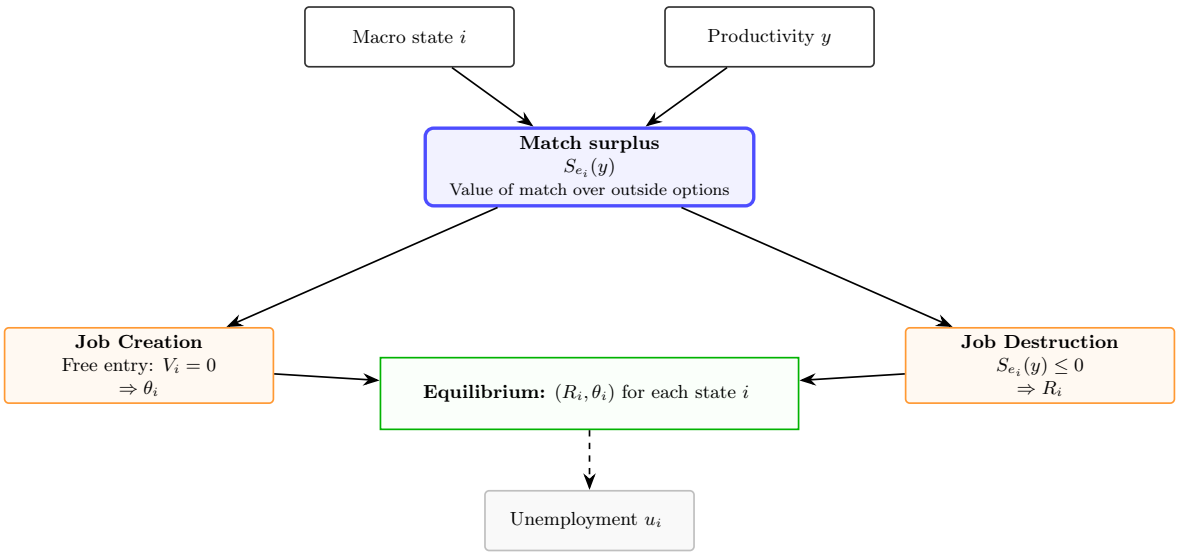


Figure 7: Surplus-based equilibrium determination

9.2.2 Match Surplus and Wage

Table 9.2.2 summarizes the model framework for match surplus definition, wage setting and consequent surplus sharing rule between firms and workers.

Concept	Expression	Interpretation
Match surplus	$S_{e_i}(y) = V_{e_i}(y) - V_{I_i} + J_{e_i}(y) - J_{V_i}$	Total gains from a worker–firm match
Wage determination	$w_{e_i}(y) = \arg\max[V_{e_i}(y) - V_{I_i}]^{\beta}[J_{e_i}(y) - J_{V_i}]^{1-\beta}$	Nash bargaining over match surplus
Worker surplus share	$V_{e_i}(y) - V_{I_i} = \beta S_{e_i}(y)$	Worker receives fraction β of surplus
Firm surplus share	$J_{e_i}(y) - J_{V_i} = (1-\beta) S_{e_i}(y)$	Firm receives remaining surplus

Using the free entry condition, Bellman equations and the sharing rule, we obtain the resulting surplus expression as a function of model parameters:

$$\left(r + \mu + \sum_{i \neq j}^{n} \Omega_{ij}\right) S_{e_i}(y) = p_i + \sigma y - \left[b_{I_i} + \theta_i \frac{\beta}{1-\beta}\gamma + \frac{\lambda}{r + \lambda + \theta_i q(\theta_i)}\left(b_{U_i} - b_{I_i}\right)\right] + \mu \mathbb{E}(S_{e_i}) + \sum_{i \neq j}^{n} \Omega_{ij} \max[S_{e_j}(y), 0] \tag{5}$$

where $\mathbb{E}[S_{e_i}] = \int \max[S_{e_i}(y'), 0]\, dG(y')$

The left-hand side represents the total flow surplus. The surplus is annuitized by the discount rate r, the job destruction idiosyncratic rate μ, and the aggregate state transition rate Ω. The match generates a flow of $p_i + \sigma y$, reduced by the opportunity cost of the worker: the value of the outside option, being unemployed insured. This includes (i) unemployment benefits while insured, (ii) the expected gain from continued job search, which in equilibrium corresponds to the second term inside the brackets and (ii) the UI exhaustion effect, which lowers the worker outside option. The term $\mu\mathbb{E}(S_{e_i})$ accounts for the expected future surplus if the job survives an idiosyncratic shock, representing the "option value" of

the specific match. Finally, the term $\sum \Omega_{ij} \max[S_{e_j}(y), 0]$ captures the macroeconomic option value of the job. It implies that if the aggregate state changes, the pair will only remain matched if the resulting surplus in the new state is positive, providing a "floor" to the match's value. It accounts for the capital gains or losses that occur if the entire economy transitions to a different aggregate state (j).

The wage equation reflects Nash bargaining between the worker and the firm, where the surplus from the match is split according to the worker's bargaining power β. Wage increases with both the worker's outside option and the productivity of the match, with the relative importance of each component governed by bargaining power.

$$w_{e_i}(y) = (1-\beta)\left[b_{I_i} + \theta_i \frac{\beta}{1-\beta}\gamma + \frac{\lambda}{r+\lambda+\theta_i q(\theta_i)}\left(b_{U_i} - b_{I_i}\right)\right] + \beta(p_i + \sigma y)$$

9.2.3 Job Creation

The job creation condition is the first equilibrium equation of the model, which is pinned down by the free entry condition and the wage bargaining sharing rule. It dictates that firms in a specific economic state i will continue to post vacancies until the marginal cost of recruitment (left hand side) is exactly offset by the firm's expected share of the match surplus (right hand side).

$$(r+\mu+\sum_{i\neq j}^{n}\Omega_{ij})\frac{\gamma}{q(\theta_i)} = (1-\beta)\left[p_i + \sigma\bar{y} + \mu\mathbb{E}(S_{e_i}) - \left[b_{I_i} + \theta_i\frac{\beta}{1-\beta}\gamma + \frac{\lambda}{r+\lambda+\theta_i q(\theta_i)}\left(b_{U_i} - b_{I_i}\right)\right]\right] + \sum_{i\neq j}^{n}\Omega_{ij}\left[\frac{\gamma}{q(\theta_j)}\right] \tag{6}$$

The left hand side represents the capitalized cost of hiring. It combines the flow cost of maintaining a vacancy (γ) with the expected duration of the search ($1/q(\theta_i)$). This is discounted by both the interest rate (r) and the idiosyncratic shock rate (μ), representing the total capital required to bring a worker into the firm. The right hand side represents the share of surplus from a match the firm gets and the expected capital gain or loss resulting from a transition between states. Firm's share of surplus includes the flow value of the job's output at creation, the option values of the job following an idiosyncratic as well as an aggregate productivity shock less the opportunity cost of employment to the worker. Finally, the expected gains or losses from changing the aggregate state of the economy reflect the variation in vacancy posting costs under different market tightness. Higher tightness implies higher vacancy posting costs.

The higher productivity levels p_i, the higher the value of a match, and therefore, the larger the

incentives for firms to post new vacancies. Therefore, from the job creation equation, we have $\theta_1 > \theta_2 > \theta_3$. Table 18 provides an interpretation macroeconomic option value of posting a vacancy in this context. Relative to a situation in which agents do not anticipate changes in macroeconomic states, this forward-looking behavior mitigates the increase in job creation during booms, as firms expect vacancy posting to become more costly as the aggregate state worsens. It also mitigates the decline in job creation during recessions, as firms' expect the situation will improve making it less costly to recruit.

State	**Term**	**Interpretation**	**Effect**
Boom (1)	$\Omega_{12}\left(\frac{\gamma}{q(\theta_2)}\right) + \Omega_{13}\left(\frac{\gamma}{q(\theta_3)}\right)$	Costs decrease as conditions worsen	Decreases job creation today
Intermediate (2)	$\Omega_{21}\left(\frac{\gamma}{q(\theta_1)}\right) + \Omega_{23}\left(\frac{\gamma}{q(\theta_3)}\right)$	Ambiguous (depends on transitions)	No clear effect
Recession (3)	$\Omega_{31}\left(\frac{\gamma}{q(\theta_1)}\right) + \Omega_{32}\left(\frac{\gamma}{q(\theta_2)}\right)$	Costs increases as conditions improve	Increases job creation today

Table 18: Anticipated vacancy posting costs across macroeconomic states

9.2.4 Job Destruction

Job destruction is the second equilibrium condition in our model, which is pinned down from $S_{e_i}(R_i) = 0$, the free entry condition and the sharing rules.

$$\begin{aligned} p_i + \sigma R_i = b_{I_i} + \theta_i \frac{\beta}{1-\beta}\gamma + \frac{\lambda}{r + \lambda + \theta_i q(\theta_i)}\left[b_{U_i} - b_{I_i}\right] \\ - \mu \mathbb{E}(S_{e_i}) - \sum_{i \neq j}^{n} \Omega_{ij} \max[S_{e_j}(R_i), 0] \end{aligned} \tag{7}$$

The left-hand side represents the minimum productivity that firms are willing to accept for a filled position. The right hand side represents the net cost of keeping the match alive. For a match to survive, its productivity must be high enough to cover these costs, which encompass the opportunity cost of employment to the worker less the option values from a job following an idiosyncratic and a macroeconomic shock. The idiosyncratic option value, represented by $\mu\mathbb{E}(S_{e_i})$, captures 'labour hoarding'. Even if the match is losing money today, the firm keeps the worker because there is a chance (μ) the worker's individual productivity will reset to a high level tomorrow. The macroeconomic option value reflects the capital gain or loss accruing to the match surplus at reservation productivity from aggregate state transitions. By maintaining a marginal match today, agents retain the option to harvest high surpluses if aggregate conditions improve (Ω_{ij}), while the max operator ensures that matches are endogenously destroyed only when the surplus cannot be restored to a positive value even

in the next best state. For a given value of θ, a higher p_i implies a lower reservation productivity. Therefore, from job destruction, we have $R_1 < R_2 < R_3$.

Table 19 summarizes how this macroeconomic option value evolves across aggregate states, highlighting that only transitions toward states in which the surplus remains non-negative contribute to survival. In particular, a marginal job in a boom (R_1) is destroyed upon any transition to worse states because its surplus becomes negative there, yielding an effective continuation value of zero. In contrast, at R_2, the job is only destroyed in the recession state. Finally, at R_3, the job survives transitions to both intermediate and boom states, as surplus is positive in those states, generating a strictly positive continuation value. This forward-looking evaluation mitigates job destruction in worsened macroeconomic conditions relative to a setting without anticipation, which would not include this option value of keeping the match in anticipation of better macroeconomic conditions.

State	**Term**	**Effective Term**	**Interpretation**
Boom (1)	$\Omega_{12}\max[S_{e_2}(R_1),0]$ $+\Omega_{13}\max[S_{e_3}(R_1),0]$	0	Match at R_1 is destroyed in states 2 and 3
Intermediate (2)	$\Omega_{21}\max[S_{e_1}(R_2),0]$ $+\Omega_{23}\max[S_{e_3}(R_2),0]$	$\Omega_{23}\max[S_{e_3}(R_2),0]$	Match at R_2 is destroyed only in recession
Recession (3)	$\Omega_{31}\max[S_{e_1}(R_3),0]$ $+\Omega_{32}\max[S_{e_2}(R_3),0]$	$\Omega_{31}\max[S_{e_1}(R_3),0]$ $+\Omega_{32}\max[S_{e_2}(R_3),0]$	Match at R_3 survives in states 1 and 2

Table 19: Marginal job survival following aggregate shocks

9.2.5 Expected Match Continuation Surplus Following Idiosyncratic Shock

Both job creation and job destruction continue to depend on the future match surplus after an idiosyncratic shock $\mathbb{E}(S_{e_i})$. In this subsection, we look at this term to better understand how a match can yield different expected surplus depending on its job-specific productivity and the aggregate state of the economy. Refer to Appendix F for the derivation of the formal expression.

As Figure 8 shows, the expected surplus from a match in a booming economy following an idiosyncratic productivity shock is higher than in an economy in recession. This follows from a higher range of productivities in which the match survives under good economic conditions. Regarding job destruction, this contributes to lower reservation productivities and, therefore, lower job destruction in booming economies. In the case of job creation, this translates into higher expected overall match surplus and, therefore, higher job creation in a boom.

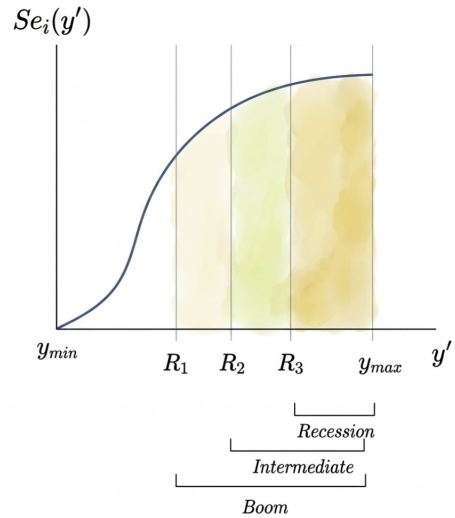


Figure 8: Expected Match Surplus After Idiosyncratic Shock

9.2.6 Equilibrium System across Macroeconomic States

We consider the three aggregate states $i = 1, 2, 3$, corresponding to a boom, a intermediate state, and a recession, with $p_1 > p_2 > p_3$. The equilibrium is characterized by the following system of job creation and job destruction equations for each state, which pins down

$$R = \begin{pmatrix} R_1 \\ R_2 \\ R_3 \end{pmatrix} \quad \theta = \begin{pmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \end{pmatrix}$$

$$
\begin{cases}
\textbf{Job creation conditions} \\
(r+\mu)\dfrac{\gamma}{q(\theta_1)} = (1-\beta)\left[p_1 + \sigma\bar{y} + \mu\mathbb{E}(S_{e_1}) - \left(b_{I_1} + \theta_1\dfrac{\beta}{1-\beta}\gamma + \dfrac{\lambda(b_{U_1}-b_{I_1})}{r+\lambda+\theta_1 q(\theta_1)}\right)\right] \\
\qquad\qquad +\Omega_{12}\left(\dfrac{\gamma}{q(\theta_2)} - \dfrac{\gamma}{q(\theta_1)}\right) + \Omega_{13}\left(\dfrac{\gamma}{q(\theta_3)} - \dfrac{\gamma}{q(\theta_1)}\right) \\
(r+\mu)\dfrac{\gamma}{q(\theta_2)} = (1-\beta)\left[p_2 + \sigma\bar{y} + \mu\mathbb{E}(S_{e_2}) - \left(b_{I_2} + \theta_2\dfrac{\beta}{1-\beta}\gamma + \dfrac{\lambda(b_{U_2}-b_{I_2})}{r+\lambda+\theta_2 q(\theta_2)}\right)\right] \\
\qquad\qquad +\Omega_{21}\left(\dfrac{\gamma}{q(\theta_1)} - \dfrac{\gamma}{q(\theta_2)}\right) + \Omega_{23}\left(\dfrac{\gamma}{q(\theta_3)} - \dfrac{\gamma}{q(\theta_2)}\right) \\
(r+\mu)\dfrac{\gamma}{q(\theta_3)} = (1-\beta)\left[p_3 + \sigma\bar{y} + \mu\mathbb{E}(S_{e_3}) - \left(b_{I_3} + \theta_3\dfrac{\beta}{1-\beta}\gamma + \dfrac{\lambda(b_{U_3}-b_{I_3})}{r+\lambda+\theta_3 q(\theta_3)}\right)\right] \\
\qquad\qquad +\Omega_{31}\left(\dfrac{\gamma}{q(\theta_1)} - \dfrac{\gamma}{q(\theta_3)}\right) + \Omega_{32}\left(\dfrac{\gamma}{q(\theta_2)} - \dfrac{\gamma}{q(\theta_3)}\right) \\
\textbf{Job destruction conditions} \\
p_1 + \sigma R_1 = b_{I_1} + \theta_1\dfrac{\beta}{1-\beta}\gamma + \dfrac{\lambda}{r+\lambda+\theta_1 q(\theta_1)}(b_{U_1} - b_{I_1}) \\
\qquad\qquad -\mu\mathbb{E}(S_{e_1}) - \Omega_{12}\max[S_{e_2}(R_1), 0] - \Omega_{13}\max[S_{e_3}(R_1), 0] \\
p_2 + \sigma R_2 = b_{I_2} + \theta_2\dfrac{\beta}{1-\beta}\gamma + \dfrac{\lambda}{r+\lambda+\theta_2 q(\theta_2)}(b_{U_2} - b_{I_2}) \\
\qquad\qquad -\mu\mathbb{E}(S_{e_2}) - \Omega_{21}\max[S_{e_1}(R_2), 0] - \Omega_{23}\max[S_{e_3}(R_2), 0] \\
p_3 + \sigma R_3 = b_{I_3} + \theta_3\dfrac{\beta}{1-\beta}\gamma + \dfrac{\lambda}{r+\lambda+\theta_3 q(\theta_3)}(b_{U_3} - b_{I_3}) \\
\qquad\qquad -\mu\mathbb{E}(S_{e_3}) - \Omega_{31}\max[S_{e_1}(R_3), 0] - \Omega_{32}\max[S_{e_2}(R_3), 0]
\end{cases}
\tag{8}
$$

9.2.7 Beveridge Curve

The model introduces new dynamics impacting the relation between vacancies and unemployment.

1. **The Matching Effect** Holding the destruction rate constant, an increase in vacancies increases market tightness θ, thereby increasing the probability of an unemployed worker finding a job. To maintain a stationary matching rate, the unemployment level must fall. This generates the classical downward-sloping segment of the Beveridge.

2. **Job Destruction Channel**: As v increases, the opportunity cost of employment increases, forcing firms to increase their reservation productivity threshold. The job destruction rate thus increases with θ. This induces higher turnover, requiring a higher unemployment level to reach a stationary state of job destruction.

The slope of the Beveridge-curve depends on the relative magnitudes of these effects. If the elas-

ticity of job destruction with respect to tightness exceeds the elasticity of the matching function, the model predicts a breakdown in the standard inverse relationship between vacancies and unemployment, potentially resulting in a positively sloped or non-monotonic Beveridge curve.

9.2.8 Asymmetries in Job Creation and Job Destruction Across the Business Cycle

In our model, firms can adjust job destruction and vacancies immediately when conditions change, meaning they are forward-looking "jump" variables. Unemployment, however, behaves very differently. It cannot jump instantly because it depends on the matching process between workers and firms, which takes time. Instead, unemployment adjusts gradually over time and is, therefore, a sticky variable.

This affects dynamics along the business cycles:

- When productivity falls (i.e., during a recession): some jobs that were just barely profitable are no longer worth keeping. Firms immediately destroy these marginal jobs and close some vacancies.
- When productivity rises (i.e., during a boom) : Firms respond by opening new vacancies right away, since hiring is more attractive. However, these vacancies take time to fill, so employment does not increase immediately.

This asymmetry is rooted in the discontinuous response of job destruction to negative aggregate productivity shocks. Transition from high productivities to low productivities induces an immediate upward shift in the reservation productivity threshold. This shift results in an instantaneous "cleansing effect": the entire stock of existing jobs whose idiosyncratic productivity components reside between the old and new reservation levels becomes economically unviable. These jobs are terminated on impact, creating a sharp spike in the job destruction rate. Because this mass-termination occurs the moment the threshold shifts, the job destruction rate dominates the job creation rate as the primary driver of unemployment at the onset of a recession.

Conversely, booms do not have a symmetric counterpart. While a higher aggregate productivity stimulates job creation by raising market tightness and incentivizing firms to hold onto jobs longer, there is no mechanism for the "instantaneous" creation of a job mass. Job creation remains a gradual process governed by the matching function and the slow expansion of the unemployment pool.

Consequently, the structural disparity between the immediate termination of marginal jobs and the gradual matching of new ones explains why the velocity of unemployment's rise during a downturn is markedly faster than its rate of decline during a subsequent boom. The job destruction rate thus imparts a specific cyclical asymmetry to the dynamic behavior of unemployment.

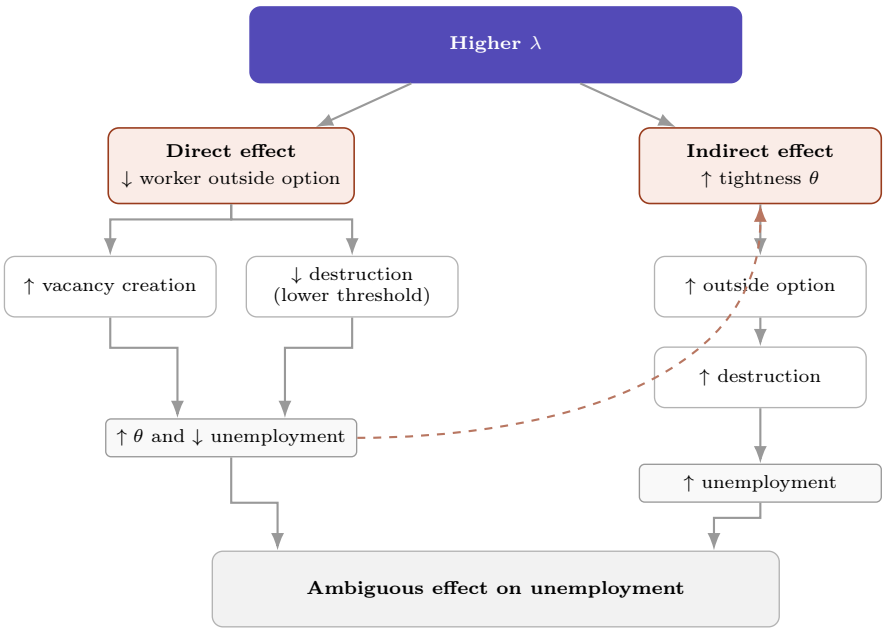


Figure 9: Reform Channels with Endogenous Job Destruction

9.3 An Illustration of Increasing Benefit Expiration Rate on Unemployment in an Artificial Economy

This section analyzes the effects of reducing potential benefit duration, in line with the 2023 reform, by increasing the benefit expiration rate within the current model. Due to time constraints, the analysis focuses on the underlying qualitative mechanisms rather than their quantitative magnitude. The section concludes by outlining directions for future work aimed at extending the framework toward a more realistic and context-relevant analysis.

9.3.1 Theoretically Ambiguous Effect of Reduced Potential Benefit Duration

An increase in the benefit expiration rate λ has theoretically ambiguous effects on equilibrium unemployment due to the presence of competing mechanisms. On the one hand, a higher λ reduces the worker's outside option in wage bargaining, which lowers the reservation productivity threshold R and induces firms to create more vacancies. This contributes to increasing market tightness θ and reducing unemployment through higher job finding rates. On the other hand, and indirectly, a higher θ increases the value of unemployment through improved job finding prospects, potentially raising the reservation productivity and inducing higher job destruction. As a result, the net effect of an increase in λ on unemployment depends on the relative strength of these opposing forces, and is therefore theoretically ambiguous. Figure 9 graphically represent these channels.

9.3.2 Model Calibration and Baseline Artificial Economy

The model is calibrated to quarterly values to simulate an artificial economy using parameters from the literature for the purpose of the illustration. These are presented in Table 20.

Parameter	**Symbol**	**Value**	**Source**
Quarterly interest rate	r	0.0125	Standard
Bargaining power	β	0.500	Hosios (5)
Matching elasticity	η	0.500	Petrongolo and Pissarides (10)
Benefit expiration hazard	λ	0.1875	Unédic (2022), 16-month avg.
Insured UI benefit	b_I	0.55	OECD replacement rate
Uninsured income	b_U	0.15	RSA + transfers, CAF (2022)
Vacancy posting cost	γ	0.40	Hagedorn and Manovskii (3)
Matching efficiency	m_0	1	Petrongolo and Pissarides (10)
Idiosyncratic shock dispersion	σ	0.3637	L'Haridon and Malherbet (6)
Idiosyncratic shock hazard rate	μ	0.08	L'Haridon and Malherbet (6)

Table 20: Externally Calibrated Model Parameters

The aggregate technology shock is modeled as a three-state Markov chain on the grid (p_1, p_2, p_3), with transition probabilities Ω_{ij} for all $i, j = 1, 2, 3$, designed to approximate an underlying AR(1) process. The parameters governing this stochastic process are taken from L'Haridon and Malherbet (6), with $\rho = 0.94$ capturing the persistence of aggregate productivity and $\nu = 0.007$ the standard deviation of innovations. In addition, the idiosyncratic component y is assumed to follow a uniform distribution, defined over the interval $[0, 2]$.

Table 21: Baseline Calibration

State-Specific Outcomes	High (1)	Medium (2)	Low (3)
Reservation productivity R	1.87	1.88	1.90
Market tightness θ	0.66	0.54	0.43
Job creation rate	0.81	0.74	0.65
Job destruction rate	0.075	0.075	0.076
Unemployment rate (%)	8.39	9.28	10.42
Insured unemployment (%)	6.82	7.4	8.09
Uninsured unemployment (%)	1.57	1.89	2.33
Aggregate Outcomes			
Aggregate unemployment (%)		9.33	
Vacancy creation rate (%)		5.02	
Separation rate (%)		7.53	
Job finding rate f		0.73	
Vacancy filling rate q		1.37	

At the aggregate level, the model delivers an unemployment rate of 9.33%, a vacancy creation rate of 5.02% and separation rate of 7.53% in steady state. Turning to state-specific outcomes, market tightness θ increases when moving toward more favorable aggregate states, reflecting improved labor market conditions and stronger vacancy creation. Reservation productivity R declines in good states, as expected, as lower job-specific productivity is required to keep matches alive.

9.3.3 Policy Experiment : Increasing Benefit Expiration Rate

An increase in the benefit expiration rate λ reduces the expected duration of unemployment benefits, thereby lowering workers' outside option in wage bargaining. This leads to a decrease in the reservation productivity threshold R, implying that lower productivity is required to keep the match alive. As a result, firms post more vacancies, increasing market tightness θ and improving job-finding prospects. Consequently, the job finding rate f increases, while the vacancy filling rate q declines due to increased competition among vacancies. Overall, these equilibrium adjustments lead to a reduction in the unemployment rate. The effect on job destruction remains limited, as the policy primarily operates through the worker's outside option rather than the productivity process itself. Finally, the composition of unemployment shifts, with a reduction in insured unemployment and a corresponding increase in uninsured unemployment, reflecting the shorter duration of benefit eligibility.

Table 22: Policy Experiment: Baseline vs. Policy Change (λ)

	Baseline	**Reform**	**Change**
Policy lever			
Benefit expiration hazard λ	0.1884	0.2512	+33.3%
Expected UI duration $D = 1/\lambda$	5.31 qtrs	3.98 qtrs	−25.0%
Labour Market Outcomes			
Unemployment rate (%)	9.33	9.00	-0.33pp
Vacancy rate (%)	5.02	4.85	-0.17pp
Separation rate (%)	7.53	7.51	-0.02pp
Matching Efficiency			
Job finding rate (f)	0.73	0.76	+0.02
Vacancy filling rate (q)	1.37	1.32	-0.05
Equilibrium Objects			
Mean tightness θ	0.54	0.58	+0.04
Mean reservation productivity R	1.88	1.87	-0.01
Unemployment Composition			
Insured unemployment (%)	7.42	6.77	-0.65pp
Uninsured unemployment (%)	1.91	2.23	+0.32pp

9.3.4 Next Steps : Toward Realistic Calibration and Reform Analysis

The next step in extending this framework consists of improving the quantitative discipline of the model by targeting a richer set of empirical moments, with a particular focus on the French labor market. While the current calibration captures key qualitative mechanisms, a more realistic assessment requires aligning the model more closely with observed data on unemployment dynamics, job flows, and labor market transitions. In addition, a natural extension concerns the discretization of the aggregate productivity process. While the current three-state Markov chain provides a tractable approximation of the underlying AR(1) process, alternative discretization techniques may offer improved accuracy. In particular, methods such as Tauchen (13) or the Rouwenhorst procedure (12) are

well suited to approximating highly persistent stochastic processes, such as the one considered here. Adopting such techniques would enhance the reliability of the quantitative results and provide a more robust foundation for the calibration of the model.

10 Conclusion

Overall, this paper develops a structural framework that links benefit duration to key labor market outcomes, including search behavior, job creation, job destruction, and the Beveridge curve. While the current analysis provides a first step toward understanding these mechanisms, future work should focus on a more rigorous empirical calibration to match key labor market moments, particularly in the French context. Further extensions of the last model would enhance its realism by incorporating endogenous search effort, minimum wage considerations, state-dependent benefit systems, and worker heterogeneity across skills or sectors. Finally, moving beyond steady-state analysis to study transition dynamics would provide a more comprehensive assessment of the policy effects, allowing for a richer evaluation of short- and medium-run adjustments to changes in benefit duration.

A Appendix : Derivation of the Job-Creation Condition

Let J denote the value of a filled job and V_v the value of an open vacancy. Firms post vacancies at flow cost γ, filled with probability $q(\theta)$.

Firm value functions

The Bellman equations for the firm are:

$$rJ = y - w + \delta(V_v - J), \tag{9}$$

$$rV_v = -\gamma + q(\theta)(J - V_v). \tag{10}$$

Free entry

In equilibrium, firms post vacancies until:

$$V_v = 0.$$

Substituting into (9):

$$(r + \delta)J = y - w \quad \Longrightarrow \quad J = \frac{y - w}{r + \delta}.$$

Substituting into (10):

$$0 = -\gamma + q(\theta)J \quad \Longrightarrow \quad \gamma = q(\theta)J.$$

Combining the two expressions yields the job-creation condition:

$$\frac{\gamma}{q(\theta)} = \frac{y - w}{r + \delta}, \tag{11}$$

which corresponds to equation (1) in the main text.

B Appendix : Derivation of the Wage Equation

Worker Bellman equations

Let (V_E, V_I, V_U) denote the value functions of employed, insured unemployed, and uninsured unemployed workers. They satisfy:

$$rV_E = w + \delta(V_I - V_E), \tag{12}$$

$$rV_I = b_I - \tfrac{1}{2}c_0 s_I^2 + s_I f(\theta)(V_E - V_I) + \lambda(V_U - V_I), \tag{13}$$

$$rV_U = b_U - \tfrac{1}{2}c_0 s_U^2 + s_U f(\theta)(V_E - V_U). \tag{14}$$

Surplus definitions

Define:

$$\Delta_I \equiv V_E - V_I, \qquad \Delta_U \equiv V_E - V_U, \qquad V_U - V_I = \Delta_I - \Delta_U.$$

Step 1 — Expression for Δ_I

Subtracting (13) from (12):

$$r(V_E - V_I) = (w - b_I) + \tfrac{1}{2}c_0 s_I^2 - \delta(V_E - V_I) - s_I f(\theta)(V_E - V_I) - \lambda(V_U - V_I).$$

Collecting terms:

$$\big(r + \delta + s_I f(\theta)\big)\Delta_I = (w - b_I) + \tfrac{1}{2}c_0 s_I^2 - \lambda(V_U - V_I). \tag{15}$$

Step 2 — Expression for $V_U - V_I$

Subtracting (14) from (13), and using $V_E - V_U = \Delta_I + (V_I - V_U)$:

$$\big(r + s_U f(\theta) + \lambda\big)(V_I - V_U) = (b_I - b_U) - \tfrac{1}{2}c_0(s_I^2 - s_U^2) + (s_I - s_U)f(\theta)\Delta_I.$$

Therefore:

$$V_U - V_I = \frac{(b_U - b_I) + \tfrac{1}{2}c_0(s_I^2 - s_U^2) - (s_I - s_U)f(\theta)\Delta_I}{r + s_U f(\theta) + \lambda}. \tag{16}$$

Step 3 — Nash bargaining

When a match is formed, the worker and the firm bargain over the wage. The total surplus of the match is:

$$S = (V_E - V_I) + (J - V_v).$$

The Nash bargaining problem is:

$$\max_{w} (V_E - V_I)^{\beta}(J - V_v)^{1-\beta}.$$

The first-order condition implies that the surplus is split according to the bargaining weights:

$$V_E - V_I = \beta S.$$

Substituting the definition of S and solving:

$$(1-\beta)(V_E - V_I) = \beta(J - V_v).$$

Under free entry $V_v = 0$, and the firm value function gives $J = \gamma/q(\theta)$. Therefore:

$$(1-\beta)\,\Delta_I = \beta\,\frac{\gamma}{q(\theta)} \quad \Longrightarrow \quad \Delta_I = \frac{\beta}{1-\beta}\,\frac{\gamma}{q(\theta)}. \tag{17}$$

Step 4 — Optimal effort

Insured workers. Maximizing V_I over s_I in (13) gives:

$$-c_0 s_I + f(\theta)\Delta_I = 0 \quad \Longrightarrow \quad \Delta_I = \frac{c_0 s_I}{f(\theta)}.$$

Substituting (17):

$$s_I = \frac{f(\theta)}{c_0}\,\frac{\beta}{1-\beta}\,\frac{\gamma}{q(\theta)}, \tag{18}$$

which corresponds to equation (iii) of system (3).

Uninsured workers. Maximizing V_U over s_U in (14) gives:

$$-c_0 s_U + f(\theta)\Delta_U = 0 \quad \Longrightarrow \quad \Delta_U = \frac{c_0 s_U}{f(\theta)}.$$

Substituting $\Delta_U = c_0 s_U / f(\theta)$ into (14):

$$rV_U = b_U + \tfrac{1}{2} c_0 s_U^2.$$

Subtracting this from (13) and using the insured FOC $s_I f(\theta)\Delta_I = c_0 s_I^2$:

$$(r+\lambda)(\Delta_U - \Delta_I) = (b_I - b_U) + \tfrac{1}{2} c_0 (s_I^2 - s_U^2).$$

Substituting (17) and $\Delta_U = c_0 s_U / f(\theta)$ yields the quadratic condition for s_U:

$$\frac{1}{2} c_0 s_U^2 + \frac{c_0(r+\lambda)}{f(\theta)} s_U + \left[(b_U - b_I) - (r+\lambda) \frac{\beta}{1-\beta} \frac{\gamma}{q(\theta)} - \frac{f(\theta)^2}{2c_0} \left(\frac{\beta}{1-\beta} \frac{\gamma}{q(\theta)} \right)^2 \right] = 0, \tag{19}$$

which corresponds to equation (iv) of system (3).

Step 5 — Wage equation

Substituting (17) into (15) and solving for w, using $V_U - V_I = \Delta_I - \Delta_U = \frac{\beta}{1-\beta} \frac{\gamma}{q(\theta)} - \frac{c_0}{f(\theta)} s_U$:

$$w = b_I - \tfrac{1}{2} c_0 s_I^2 + \big(r + \delta + s_I f(\theta) + \lambda\big) \frac{\beta}{1-\beta} \frac{\gamma}{q(\theta)} - \lambda \frac{c_0}{f(\theta)} s_U, \tag{20}$$

which corresponds to equation (ii) of system (3).

C Appendix : Derivation of the Effort Equations under the Minimum Wage

The job-creation condition is derived exactly as in Appendix A, replacing the Nash wage w^* with the floor w_0. This appendix derives equations (ii) and (iii) of system (4).

Starting point

The worker Bellman equations (12)–(14) and the first-order conditions for effort are unchanged. In particular, Steps 1 and 2 of Appendix B remain valid with $w = w_0$ and $\theta = \theta_0$: equations (15) and (16) hold. The Nash sharing rule (17), however, does not: the wage is fixed at w_0 rather than determined by bargaining, so Δ_I is no longer pinned by (17). The substitution performed in Steps 4 and 5 of Appendix B is therefore unavailable.

Effort equations without Nash substitution

The first-order conditions still give:

$$\Delta_I = \frac{c_0 s_I}{f(\theta_0)}, \qquad \Delta_U = \frac{c_0 s_U}{f(\theta_0)}.$$

These are the only restrictions on Δ_I and Δ_U available in this regime.

Equation for s_I. Substituting $\Delta_I = c_0 s_I / f(\theta_0)$ and $V_U - V_I = \Delta_I - \Delta_U = c_0(s_I - s_U)/f(\theta_0)$ into (15) with $w = w_0$:

$$\big(r + \delta + s_I f(\theta_0)\big)\frac{c_0 s_I}{f(\theta_0)} = (w_0 - b_I) + \tfrac{1}{2}c_0 s_I^2 - \lambda\,\frac{c_0(s_I - s_U)}{f(\theta_0)}.$$

Moving the λ term to the left:

$$\big(r + \delta + s_I f(\theta_0) + \lambda\big)\frac{c_0\, s_I}{f(\theta_0)} = (w_0 - b_I) + \tfrac{1}{2}c_0 s_I^2 + \lambda\,\frac{c_0\, s_U}{f(\theta_0)}.$$

Expanding the term $s_I f(\theta_0) \cdot c_0 s_I / f(\theta_0) = c_0 s_I^2$ on the left and moving it to the right:

$$(r + \delta + \lambda)\,\frac{c_0\, s_I}{f(\theta_0)} = (w_0 - b_I) - \tfrac{1}{2}c_0 s_I^2 + \lambda\,\frac{c_0\, s_U}{f(\theta_0)}, \tag{21}$$

which corresponds to equation (ii) of system (4).

Equation for s_U. The relation:

$$(r + \lambda)(\Delta_U - \Delta_I) = (b_I - b_U) + \tfrac{1}{2}c_0(s_I^2 - s_U^2)$$

is derived in Step 4 of Appendix B solely from the Bellman equations and the first-order conditions, without any use of Nash. It therefore remains valid here. Substituting $\Delta_I = c_0 s_I / f(\theta_0)$ and $\Delta_U = c_0 s_U / f(\theta_0)$:

$$(r + \lambda)\,\frac{c_0(s_U - s_I)}{f(\theta_0)} = (b_I - b_U) + \tfrac{1}{2}c_0\big(s_I^2 - s_U^2\big), \tag{22}$$

which corresponds to equation (iii) of system (4).

D Appendix: Match Surplus and Wage Determination

This appendix provides a detailed derivation of the match surplus and wage equations starting from the Bellman conditions. The goal is to make transparent how the key equilibrium objects used in the main text—match surplus, the worker's outside option, and wages—emerge from the primitives of the model.

We proceed in four steps. First, we present the relevant Bellman equations for firms and workers. Second, we combine these value functions to construct the match surplus and derive its dynamic equation. Third, we characterize the worker's outside option by solving the unemployment value equations. Finally, we use Nash bargaining to derive wages as a function of surplus.

Throughout, intermediate steps are reported explicitly to guide the reader through the derivations.

1. Bellman Equations

Firm (vacancy)

$$rJ_{V_i} = -\gamma + q(\theta)\left[J_{e_i}(\bar{y}) - J_{V_i}\right] + \sum_{i \neq j}^{n} \Omega_{ij}\left[\Pi_{V_j} - J_{V_i}\right]$$

Firm (filled job)

$$\begin{aligned} rJ_{e_i}(y) = p_i + \sigma y - w_i(y) \\ + \mu\left[\int \max\left[J_{e_i}(y'), J_{V_i}\right] dG(y') - J_{e_i}(y)\right] \\ + \sum_{i \neq j}^{n} \Omega_{ij}\left[\max\left[J_{e_j}(y), V_j\right] - J_{e_i}(y)\right] \end{aligned}$$

Worker (employed)

$$\begin{aligned} rV_{e_i}(y) = w_{e_i}(y) + \mu\left[\int \max\left[V_{e_i}(y'), V_{I_i}\right] dG(y') - V_{e_i}(y)\right] \\ + \sum_{i \neq j}^{n} \Omega_{ij}\left[\max\left[V_{e_j}(y), V_{I_i}\right] - V_{e_i}(y)\right] \end{aligned}$$

Worker (unemployed, insured)

$$\begin{aligned} rV_{I_i} = b_I + \theta_i q(\theta_i)\left(V_{e_i}(\bar{y}) - V_{I_i}\right) + \lambda(V_{U_i} - V_{I_i}) \\ + \sum_{i \neq j}^{n} \Omega_{ij}\left[V_{I_j} - V_{I_i}\right] \end{aligned}$$

Worker (unemployed, uninsured)

$$rV_{U_i} = b_U + \theta_i q(\theta_i)\big(V_{e_i}(\bar{y}) - V_{U_i}\big) + \sum_{j \neq i} \Omega_{ij}(V_{U_j} - V_{U_i})$$

—

2. Match Surplus

We define the match surplus as the joint value of the match relative to unemployment:

$$S_{e_i}(y) = V_{e_i}(y) - V_{I_i} + J_{e_i}(y) - J_{V_i}$$

Using free entry ($J_{V_i} = 0$) and combining the Bellman equations, we obtain:

Surplus equation

$$\left(r + \mu + \sum_{i \neq j}^{n} \Omega_{ij}\right) S_{e_i}(y) = p_i + \sigma y - rV_{I_i} + \mu \mathbb{E}[S_{e_i}] + \sum_{i \neq j}^{n} \Omega_{ij} \left[V_{I_j} - V_{I_i}\right] + \sum_{i \neq j}^{n} \Omega_{ij} S_{e_j}(y)$$

where $\mathbb{E}[S_{e_i}] = \int \max[S_{e_i}(y'), 0]\, dG(y')$

—

3. Worker Outside Option

To express the surplus in a tractable way, we solve for the worker's outside option.

Step 1: Difference between unemployment states

$$V_{U_i} - V_{I_i} = \frac{b_U - b_I}{r + \lambda + \theta_i q(\theta_i)}$$

Step 2: Substitute into insured value Using free entry and sharing rule:

$$S_{e_i}(\bar{y}) = \frac{\gamma}{q(\theta_i)(1 - \beta)}$$

we obtain:

$$\boxed{rV_{I_i} - \sum_{j \neq i} \Omega_{ij}(V_{I_j} - V_{I_i}) = b_I + \theta_i \frac{\beta}{1 - \beta}\gamma + \frac{\lambda}{r + \lambda + \theta_i q(\theta_i)}(b_U - b_I)}$$

This motivates defining the **effective outside option**:

$$\mathcal{B}_i = b_I + \theta_i \frac{\beta}{1-\beta}\gamma + \frac{\lambda}{r + \lambda + \theta_i q(\theta_i)}(b_U - b_I)$$

—

4. Wage Determination

Wages are determined by Nash bargaining:

$$w_{e_i}(y) = \text{argmax}\, [V_{e_i}(y) - V_{I_i}]^{\beta} \, [J_{e_i}(y) - J_{V_i}]^{1-\beta}$$

Sharing rule

$$V_{e_i}(y) - V_{I_i} = \beta S_{e_i}(y), \quad J_{e_i}(y) = (1-\beta) S_{e_i}(y)$$

Wage equation

$$\boxed{w_i(y) = (1-\beta)\,\mathcal{B}_i + \beta(p_i + \sigma y)}$$

Interpretation. Wages are a convex combination of:

- the worker's outside option $\mathcal{B}_i$,
- current match productivity $p_i + \sigma y$.

—

5. Final Surplus Equation

Substituting the outside option into the surplus equation yields:

$$\boxed{\begin{aligned}(r + \mu + \sum_{j \neq i} \Omega_{ij}) S_{e_i}(y) = p_i + \sigma y \\ - \mathcal{B}_i + \mu \mathbb{E}[S_{e_i}] \\ + \sum_{j \neq i} \Omega_{ij} \max\{S_{e_j}(y), 0\}\end{aligned}}$$

This expression summarizes the determination of match surplus and serves as the key input into the job creation and job destruction conditions in the main text. Appendix F derives the expresions for expected continuation surplus after an idiosyncratic productivity shock.

E Appendix: Model 2 Steady-State Equilibrium Conditions

Overview. This appendix summarizes the steady-state equilibrium conditions for the model with endogenous job destruction. We present the job creation, job destruction, and unemployment equations in a compact form, emphasizing the role of expected continuation values and outside options.

Job Creation

Free entry implies:

$$\frac{\gamma}{q(\theta_i)} = (1-\beta)S_{e_i}(\bar{y}) \tag{23}$$

Using the surplus equation evaluated at $\bar{y}$, job creation can be written as:

$$\begin{aligned}(r+\mu)\frac{\gamma}{q(\theta_i)} = (1-\beta)\Bigg[p_i + \sigma\bar{y} + \mu\mathbb{E}(S_{e_i}) - \left(b_I + \theta_i\frac{\beta}{1-\beta}\gamma + \frac{\lambda}{r+\lambda+\theta_i q(\theta_i)}\left(b_U - b_I\right)\right)\Bigg] \\ + \sum_{i\neq j}^{n}\Omega_{ij}\left[\frac{\gamma}{q(\theta_j)} - \frac{\gamma}{q(\theta_i)}\right]\end{aligned}$$

Job Destruction

The reservation productivity R_i is defined by $S_{e_i}(R_i) = 0$. Evaluating the surplus equation at R_i yields:

$$\begin{aligned}p_i + \sigma R_i = b_I + \theta_i\frac{\beta}{1-\beta}\gamma + \frac{\lambda}{r+\lambda+\theta_i q(\theta_i)}\left[b_U - b_I\right] - \mu\mathbb{E}(S_{e_i}) \\ - \sum_{i\neq j}^{n}\Omega_{ij}\max[S_{e_j}(R_i), 0]\end{aligned}$$

Unemployment

Law of motion

$$\dot{u}_i = \mu G(R_i)(1-u_i) - \theta_i q(\theta_i)u_i \tag{24}$$

Steady state

$$u_i = \frac{\mu G(R_i)}{\mu G(R_i) + \theta_i q(\theta_i)} \tag{25}$$

F Appendix : Model 2 Expected Continuation Surplus after Idiosyncratic Productivity Shock and Fully Specified Steady State Equilibrium

Overview. This appendix derives the expected surplus arising from idiosyncratic productivity shocks in the model with endogenous job destruction and provides the fully specified steady-state equilibrium conditions. We proceed in three steps.

First, we characterize the surplus functions $S_{e_i}(y)$ for each aggregate state $i \in \{1, 2, 3\}$ as piecewise linear functions of idiosyncratic productivity, highlighting the role of endogenous thresholds (R_1, R_2, R_3).

Second, we derive closed-form expressions for the expected surplus $\mathbb{E}[S_{e_i}]$ using integration by parts and show how these can be written as weighted sums of tail probabilities of the productivity distribution.

Finally, we use these results to express job creation and job destruction conditions in terms of two sufficient statistics: the expected continuation surplus and the effective outside option. This yields a compact representation of the steady-state equilibrium across aggregate states.

Productivity States

Let $y \in [\underline{y}, \bar{y}]$ denote the idiosyncratic productivity of a match. For expositional clarity, we define three aggregate productivity states:

- **State 1: Boom (Good)** with productivity p_1
- **State 2: Intermediate (Medium)** with productivity p_2
- **State 3: Recession (Low)** with productivity p_3

We assume $p_1 > p_2 > p_3$, so that higher states correspond to better economic conditions. The idiosyncratic productivity y fluctuates within these states according to the distribution $y \sim G(y)$, with $y \in [\underline{y}, \bar{y}]$.

Piecewise Expressions of $S_{e_i}(y)$

State	Interval of y	Expression for $S_{e_i}(y)$
High (1)	$[\underline{y}, R_1]$	0
	$[R_1, R_2]$	$\frac{1}{r+\mu+\Omega_{12}+\Omega_{13}}\Big[p_1+\sigma y-\big(b_{I_1}+\theta_1\frac{\beta}{1-\beta}\gamma+\frac{\lambda}{r+\lambda+\theta_1 q(\theta_1)}(b_{U_1}-b_{I_1})\big)+\mu\int_{R_1}^{\bar{y}} S_{e_1}(y')dG(y')\Big]$
	$[R_2, R_3]$	$\frac{1}{r+\mu+\Omega_{12}+\Omega_{13}}\Big[p_1+\sigma y-\big(b_{I_1}+\theta_1\frac{\beta}{1-\beta}\gamma+\frac{\lambda}{r+\lambda+\theta_1 q(\theta_1)}(b_{U_1}-b_{I_1})\big)+\mu\int_{R_1}^{\bar{y}} S_{e_1}(y')dG(y')+\Omega_{12}S_{e_2}(y)\Big]$
	$[R_3, \bar{y}]$	$\frac{1}{r+\mu+\Omega_{12}+\Omega_{13}}\Big[p_1+\sigma y-\big(b_{I_1}+\theta_1\frac{\beta}{1-\beta}\gamma+\frac{\lambda}{r+\lambda+\theta_1 q(\theta_1)}(b_{U_1}-b_{I_1})\big)+\mu\int_{R_1}^{\bar{y}} S_{e_1}(y')dG(y')+\Omega_{13}S_{e_3}(y)\Big]$
Medium (2)	$[\underline{y}, R_2]$	0
	$[R_2, R_3]$	$\frac{1}{r+\mu+\Omega_{21}+\Omega_{23}}\Big[p_2+\sigma y-(b_{I_2}+\theta_2\frac{\beta}{1-\beta}\gamma+\frac{\lambda}{r+\lambda+\theta_2 q(\theta_2)}(b_{U_2}-b_{I_2}))+\mu\int_{R_2}^{\bar{y}} S_{e_2}(y')dG(y')+\Omega_{21}S_{e_1}(y)\Big]$
	$[R_3, \bar{y}]$	$\frac{1}{r+\mu+\Omega_{21}+\Omega_{23}}\Big[p_2+\sigma y-(b_{I_2}+\theta_2\frac{\beta}{1-\beta}\gamma+\frac{\lambda}{r+\lambda+\theta_2 q(\theta_2)}(b_{U_2}-b_{I_2}))+\mu\int_{R_2}^{\bar{y}} S_{e_2}(y')dG(y')+\Omega_{21}S_{e_1}(y)+\Omega_{23}S_{e_3}(y)\Big]$
Low (3)	$[\underline{y}, R_3]$	0
	$[R_3, \bar{y}]$	$\frac{1}{r+\mu+\Omega_{31}+\Omega_{32}}\Big[p_3+\sigma y-(b_{I_3}+\theta_3\frac{\beta}{1-\beta}\gamma+\frac{\lambda}{r+\lambda+\theta_3 q(\theta_3)}(b_{U_3}-b_{I_3}))+\mu\int_{R_3}^{\bar{y}} S_{e_3}(y')dG(y')+\Omega_{31}S_{e_1}(y)+\Omega_{32}S_{e_2}(y)\Big]$

Expected Surplus $\mathbb{E}[S_{e_i}]$ and Slopes

Expected Value

The expected surplus is:

$$\mathbb{E}[S_{e_i}] = \int_{R_i}^{\bar{y}} S_{e_i}(y')g(y')dy'$$

Integration by parts:

$$\int_{R_i}^{\bar{y}} S_{e_i}(y')g(y')dy' = \underbrace{S_{e_i}(\bar{y})}_{\text{boundary term}} - \int_{R_i}^{\bar{y}} S'_{e_i}(y')G(y')dy' = \frac{\partial S_{e_i}}{\partial x}\int_{R_i}^{\bar{y}}[1-G(y')]dy'$$

Piecewise Derivatives Across Regions

Define the derivative vector $\mathbf{S}'(y) = (\partial S_{e_1}/\partial y, \partial S_{e_2}/\partial y, \partial S_{e_3}/\partial y)^T$. The system of linear equations is:

Region	State	Linear System ($A\mathbf{S}' = b$)
$y \in [R_3, \bar{y}]$	High	$(r+\mu+\Omega_{12}+\Omega_{13})S'_1 - \Omega_{12}S'_2 - \Omega_{13}S'_3 = \sigma$
	Medium	$-\Omega_{21}S'_1 + (r+\mu+\Omega_{21}+\Omega_{23})S'_2 - \Omega_{23}S'_3 = \sigma$
	Low	$-\Omega_{31}S'_1 - \Omega_{32}S'_2 + (r+\mu+\Omega_{31}+\Omega_{32})S'_3 = \sigma$
$y \in [R_2, R_3]$	High	$(r+\mu+\Omega_{12}+\Omega_{13})S'_1 - \Omega_{12}S'_2 = \sigma$
	Medium	$-\Omega_{21}S'_1 + (r+\mu+\Omega_{21}+\Omega_{23})S'_2 = \sigma$
	Low	$(r+\mu+\Omega_{31}+\Omega_{32})S'_3 = 0$
$y \in [R_1, R_2]$	High	$(r+\mu+\Omega_{12}+\Omega_{13})S'_1 = \sigma$
	Medium & Low	$S'_2 = S'_3 = 0$
$y \in [\underline{y}, R_1]$	All States	$S'_1 = S'_2 = S'_3 = 0$

Compact Matrix Form

For each region, define a matrix A such that:

$$A\mathbf{S}' = b, \quad \text{with } b_i = \begin{cases} \sigma, & S_i \text{ active} \\ 0, & \text{otherwise} \end{cases}$$

Example: High region $y \in [R_3, \bar{y}]$: $\begin{pmatrix} r+\mu+\Omega_{12}+\Omega_{13} & -\Omega_{12} & -\Omega_{13} \\ -\Omega_{21} & r+\mu+\Omega_{21}+\Omega_{23} & -\Omega_{23} \\ -\Omega_{31} & -\Omega_{32} & r+\mu+\Omega_{31}+\Omega_{32} \end{pmatrix} \begin{pmatrix} S_1' \\ S_2' \\ S_3' \end{pmatrix} = \begin{pmatrix} \sigma \\ \sigma \\ \sigma \end{pmatrix}$

Expected Surplus in Terms of Slopes

Define slope coefficients b_{ij} for each interval $[R_j, R_{j+1}]$. Then:

$$\mathbb{E}[S_{e_1}] = b_{11} \int_{R_3}^{\bar{y}} [1 - G(y')]dy' + b_{12} \int_{R_2}^{R_3} [1 - G(y')]dy' + b_{13} \int_{R_1}^{R_2} [1 - G(y')]dy'$$
$$\mathbb{E}[S_{e_2}] = b_{21} \int_{R_3}^{\bar{y}} [1 - G(y')]dy' + b_{22} \int_{R_2}^{R_3} [1 - G(y')]dy'$$
$$\mathbb{E}[S_{e_3}] = b_{31} \int_{R_3}^{\bar{y}} [1 - G(y')]dy'$$

Fully Specified Steady-State

Job Creation

For each state $i \in \{1, 2, 3\}$, job creation satisfies:

$$(r+\mu)\frac{\gamma}{q(\theta_i)} = (1-\beta)\Big[p_i + \sigma\bar{y} + \mu\mathcal{E}_i - \mathcal{B}_i\Big] + \sum_{j\neq i}\Omega_{ij}\left(\frac{\gamma}{q(\theta_j)} - \frac{\gamma}{q(\theta_i)}\right) \tag{26}$$

Job Destruction (All States)

$$p_i + \sigma R_i = \mathcal{B}_i - \mu\mathcal{E}_i - \sum_{j\neq i}\Omega_{ij}\max(S_{e_j}(R_i), 0)$$

Expected continuation surplus.

$$\mathcal{E}_i \equiv \sum_{j=i}^{3} b_{ij}\,\Psi(R_j, R_{j+1}), \quad \text{where} \quad \Psi(a,b) \equiv \int_a^b (1-G(y'))\,dy' \tag{27}$$

Effective outside option.

$$\mathcal{B}_i \equiv b_{I_i} + \theta_i\frac{\beta}{1-\beta}\gamma + \frac{\lambda}{r+\lambda+\theta_i q(\theta_i)}(b_{U_i} - b_{I_i}) \tag{28}$$

Simplifications by region ordering:

- State 1: all continuation terms are zero
- State 2: only $S_{e1}(R_2) > 0$
- State 3: both $S_{e1}(R_3), S_{e2}(R_3) > 0$

Unemployment (All States)

Law of motion:

$$\dot{u}_i = \mu G(R_i)(1-u_i) - \theta_i q(\theta_i) u_i$$

Steady state:

$$u_i = \frac{\mu G(R_i)}{\mu G(R_i) + \theta_i q(\theta_i)}$$

Decomposition :

$$u_i^I = \frac{\mu G(R_i)(1-u_i)}{\theta_i q(\theta_i) + \lambda}, \quad u_i^U = \frac{\lambda}{\theta_i q(\theta_i)} u_i^I$$

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I Comparative table of academic contributions

Table 23: Comparative table of the relevant academic contributions on unemployment insurance design

Paper	Model	Distinctive Features	Quantitative and macro findings
Marinescu & Skandalis (2019)	Job search model, Mortensen (1997) with endogenous search intensity and finite UI benefit duration (partial equilibrium & French administrative data)	Assumes directed search, modeling benefit exhaustion & eligibility criterion	Job finding rate increases by ∼60% right before benefit exhaustion. UI mainly shifts timing of search effort (= temporal substitution effects).
Brébion, Briole & Khoury (2025)	Dynamic search model (Morten-Pissarides)	Studies eligibility thresholds exploiting the 2009 French; firm *ex ante* vacancy adjustment; labor market segmentation	Entitlement extensions (i) affect employment duration and (ii) induce vacancy posting responses. They document a strong entitlement effect[1] and significant spillovers effects. Suggests equilibrium vacancy channel, but not a fully solved GE model.
Landais, Michaillat & Saez (2018)	Dynamic general equilibrium search-and-matching model (Mortensen–Pissarides)	Study aggregate demand effects with congestion externalities; endogenous tightness; Baily-Chetty replacement rate + counter cyclical correction term; features unlimited UI	Optimal UI is counter cyclical once tightness externalities are internalized. Macro elasticity of unemployment is smaller than micro elasticity due to vacancy adjustment. In other words, when aggregate demand is low, unemployment is inefficiently high because there are too few vacancies
Moyen & Stähler (2014)	Dynamic stochastic GE model with search-and-matching frictional labor market and wage rigidity	Institutional heterogeneity; business cycle shocks; endogenous job creation	Counter cyclical might be UI optimal in flexible labor markets (USA) but not necessarily in rigid European settings. Generous UI slows adjustment when wages are rigid.
Mitman & Rabinovich (2015)	Equilibrium search and matching model (Diamond Mortensen Pissarides) with aggregate productivity shocks	Pro cyclical optimal benefit level; endogenous job creation and search effort; heterogenous agents & risk averse workers; business cycles aggregate labor productivity shocks	Explores GE effect on optimal cyclical modulation; find that optimal benefits display mildly pro cyclical pattern.
Friedriksson & Holmlund (1998)	Equilibrium search and matching model, Pissarides (1990) with endogenous wage bargaining and search effort	Free entry; wage bargaining; endogenous search effort; two benefit levels; no savings	Higher benefits raise negotiated wages via outside option effect, reduce vacancy creation and increase equilibrium unemployment; optimal insurance should provide declining benefits profiles over unemployment spell.
Cahuc & Lehmann (2000)	Discrete job search model with endogenous job search effort and wage determination	Rawlsian welfare criterion for aggregation of preferences; declining time sequence of unemployment benefit; three state model for workers: employed, short-term unemployed, long-term unemployed	Declining benefits may reduce welfare of long term unemployed. Optimal design depends strongly on institutional rigidity.
Joseph & Maingé (2023)	Continuous dynamic equilibrium search model, Mortensen (1977) with endogenous job arrival rates and job quality	Finite duration of UI benefits generating a non stationary environment: benefits vary over unemployment spell; exogenous wage and tax; risk-averse workers; budget constraint is to be balanced in average but not at every time.	Optimal duration is finite and positive given the non stationary condition incurred by the agency problem.
Rioux & Ortega (2002)	Extended search-and-matching model from Pissarides (1990) calibrated to France	Segmented labor market with two unemployment status: workers perceiving unemployment insurance or unemployment assistance; Nash bargaining	Policies softening UI eligibility slightly reduce unemployment. A change in compensation has not the same quantitative impact on employment and welfare in a country where both UI and UA coexist vs countries where only one exists
Rioux & Ortega (2010)	Equilibrium matching model, Pissarides (2000) calibrated to the France, Denmark and the US	Re-entitlement effects; non automatic eligibility rules; two-tier unemployment compensation; Nash bargaining; wage differential	Re-entitlement mechanism significantly mitigates wage and unemployment responses for all countries considered. Optimal compensation features declining benefits and eligibility requirements.
Schmieder, von Wachter & Bender (2012)	Job search model with liquidity constraint and endogenous search intensity on German data	State-dependent disincentive effects; welfare gains and business cycles, Chetty (2008)	Find modest effects of a one month extension in UI on unemployment duration and slightly smaller effects during recessions. Support welfare formula with weakly declining unemployment effects and strongly counter cyclical exhaustion rates for countries with constant UI duration over business cycles

Notes: [1] Separation rates spike sharply right after workers become eligible while they also tend to remain in relatively low-quality jobs long enough to secure UI rights. Overall, UI eligibility improves job mobility and match quality

J List of abbreviations

BIT Bureau international du travail (French name for ILO)

CAF Caisse d'allocations familiales

DARES Direction de l'animation de la recherche, des études et des statistiques (French Ministry of Labor)

GE General equilibrium

ILO International labour organization

INSEE Institut national de la statistique et des études économiques

IZA Institute of labor economics

LMS Labor market search (model)

MPC Marginal propensity to consume

OECD Organisation for economic co-operation and development

PBD Potential benefits duration

RSA Revenu de solidarité active

UI Unemployment insurance

K List of references

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